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A cross-ethnic 8-gene RAI-lineage transcriptomic readout (TF_3 ⊕ $effector_5$) stratifies fusion-driven, epigenetically silenced dark matter in BRAF/RAS-negative thyroid cancer across Korean and Western cohorts

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Running title: Cross-ethnic RAI-lineage panel in thyroid dark matter · **Keywords:** papillary thyroid cancer; BRAF/RAS-negative; RAI refractoriness; cross-ethnic transcriptomic panel; tyrosine-kinase fusion; promoter hypermethylation; TF_3 (FOXE1/NKX2-1/PAX8); RAI_8; Korean cohort; selection-layer biomarker.

Contents.

Pre-submission verification status (2026-05-20). This draft is internally self-audited against `manuscript_v8` audit-locked values. **Audit-locked numbers (verified):** cohorts (TCGA n=504, MSK n=117, K2 n=260 sequenced / n=235 Paper-1 cohort post-QC, Lee 2024 n=630 post-QC, Mun 2025 n=336, GSE286332 n=18); DM1 axis (TPO HM450 d=2.30; mean 8-gene β 0.385 vs 0.253; ARI 0.49/0.90/0.92; Driver_anchor ARI −0.007; BRAF V600E vs WT mRNA d=−0.04 NS); fusion (76.8% DM1 fusion+, OR 7.41 [4.32, 12.74], RET-fusion capture 27/33 = 81.8%, MAR χ^2 p = 0.56, sensitivity OR 7.18–9.07); survival (TCGA HR 2.30 [0.77, 6.88], MSK HR 2.67 [1.17, 6.10], pooled HR 2.53 [1.31, 4.89], I^2=0%); cross-cohort (TF_3 K2 FULL AUC 0.738 + I^2 35.9%; $NONOVERLAP_8$ ComBat pooled 0.700; RAI_8 RBF-SVM 5-fold CV 0.963 / LogReg 0.955 / quantum-kernel 0.369; K2 balanced n=34 RAI_8 AUC 0.275 label-transferability disclosure); MAPK × Panel-8 cross-cohort Spearman ρ = −0.327 [−0.376, −0.278], n=1,287, I^2 = 72.9% (Q14); cluster stability 97.4%; permutation null p = 0.0002; thyroid-matched bio-prior null p = 0.007; label-permutation panel-conditional null $p < 10^{-4}$; knock-out n−1 floor d = 1.54 (DIO1); immune residualization 56% retention (raw d=1.78 → d=1.00); deconvolution retention 24/24/70/47% (NNLS / Ridge-NNLS / LR-clip / nu-SVR). **Pending author-keyboard items:** §1.1 Hook, §3.1 Discussion opening (Landa 2016 cite correction), §3.4 Limitations, Q9 (reviewer hardest question), corresponding-author institutional email, ORCID IDs, Zenodo DOI mint at submission, GitHub release tag. **Reviewer-driven additions 2026-05-20 (M1–M6 revision):** (i) **Driver-stratified DM1 prevalence recomputed from source TSV** (`d4p2_tcga_hashimoto_signature/tcga_with_clinical_mutations.tsv`): BRAF V600E+ DM1 0.7% (2/273); BRAF/RAS-neg DM1 49.1% (85/173); RAS+ DM1 96.4% (53/55); OR vs driver-positive 4.78 [3.15, 7.24], Fisher p = 6.5×10^{-14}. (ii) **Within-K2 unsupervised k=2 KMeans on the 8-gene panel (n=260):** Cohen's d = 1.94, Korean DM1 prevalence 60.8% (158/260), TCGA-trained continuous transfer probability vs within-K2 unsupervised label AUC = 0.883 — the K2 balanced n=34 transfer-label AUC 0.275 is therefore identified as calibration drift, not axis failure. (ii-bis) **Within-Lee 2024 (GSE213647, n=632) independent Korean cohort unsupervised replication:** 7-of-8 panel genes recovered, multidim k=2 KMeans Cohen's d = 5.93 (all-cohort) / 5.48 (tumour-only n=370); silenced cluster correctly captures advanced disease (3/8 UTC/ATC, 1/9 PDFP). (ii-tris) **Within-Mun 2025 protein (n=336) cross-modality unsupervised replication added 2026-05-21:** 7-of-8 panel proteins (NKX2-1 absent), multidim k=2 KMeans Cohen's d = 2.86; ATC 108/113 (95.6%) and PDTC 31/46 (67.4%) → DM1 cluster, monotonic ATC→PDTC→PTC phenotype gradient. Cross-modality RNA + protein replication closed. (ii-quater) **Cross-cohort master forest added (Figure 11b, 2026-05-21):** 12 entries across 9 distinct cohorts × 3 modalities, all direction-consistent, mean Cohen's d = 2.89, median = 2.37, min = 1.56. (iii) **OS / PFI framing honestly corrected:** directly computed multivariable PFI Cox PH on TCGA-THCA (n=432, 27 events, lifelines 0.30) yields **DM1 HR_{PFI} = 0.78 [0.18, 3.48], p = 0.74** (multivariate) and HR_{PFI} = 1.11 [0.47, 2.62], p = 0.82 (univariate) — *not statistically significant*; primary-PTC DM1 prognostic signal is underpowered in TCGA. OS HR 2.30 [0.77, 6.88] in TCGA is also non-significant. The pooled OS HR 2.53 [1.31, 4.89] is driven by MSK-IMPACT advanced disease (HR 2.67 [1.17, 6.10], 38 events). Primary-PTC prognostic claims reserved for prospective Bundang cohort. (iv) **NRI methodology disclosed** (continuous-NRI per Pencina 2008, 2,000 stratified bootstrap iterations; TDS-16 NRI +0.777 explicitly acknowledged as a defensible alternative main-panel architecture in Discussion 3.2). (v) **Cross-ethnic claim is now explicitly anchored on within-K2 unsupervised replication + GSE286332 ground-truth + ComBat-pooled + transfer-probability ranking**, with planned prospective Bundang validation referenced in the abstract conclusion. **Previously removed and unchanged:** Q5 Little's MCAR (replaced with chi^2 MAR p = 0.56), Q7 CIBERSORTx specific numbers (replaced with dual-residualization 56% retention + four-method deconvolution 24/24/70/47%), HM450 within-DM1 fusion+/− Δβ=0.04 (replaced with cell-composition |d|≤0.42 argument).

Abstract

Background. Approximately 23% of papillary thyroid carcinomas (PTC) lack *BRAF* or *RAS* driver mutations and constitute a clinically heterogeneous "dark matter" compartment in which radioiodine (RAI) treatment decisions remain mechanistically unguided. Whether a compact transcriptomic readout can stratify this compartment *across ethnic backgrounds* — the determining property for clinical deployment — has not been established. **Methods.** We re-frame and re-validate the 8-gene RAI-lineage panel (RAI_8: *SLC5A5/NIS, TPO, TG, TSHR, PAX8, NKX2-1, FOXE1, DIO1*) and its 3-gene cross-ethnic anchor (TF_3: *FOXE1, NKX2-1, PAX8*) in TCGA-THCA (n=504), MSK-IMPACT (n=117), Korean K2/PRJEB11591 (n=260 RNA-seq), Korean GSE286332 (n=18 full-transcriptome), Lee 2024/GSE213647 (n=630), Mun 2025 proteogenomics (n=336), and four external GPL570 cohorts (GSE29265, GSE33630, GSE53157, GSE65144; n=287). Discovery uses pan-genome top-5000 MAD KMeans, TIERA67 categorical anchor, RandomForest importance on a 55-gene driver-excluded clean pool, supervised LogReg / RBF-SVM and quantum-kernel SVM comparators, ComBat batch correction (Johnson 2007), DerSimonian-Laird and Han-Eskin RE2 meta-analysis, Stouffer Z, leave-one-cohort-out (LOCO) sensitivity, 5,000-iteration permutation null, label-permutation panel-conditional null, thyroid-matched bio-prior null, knock-out n-1 floor, NRI/IDI vs canonical reference, and Vickers decision-curve analysis. **Results.** The DM1/DM2 axis defined by TIERA67 is reproduced by unbiased pan-genome top-5000 (ARI 0.92 vs 0.90; hypergeometric $p=3\times10^{-4}$ for TIERA67 enrichment in pan-genome top-100). The 8-gene panel reads out the same axis (within-cohort z, TCGA 5-fold CV AUC = 0.861; LogReg AUC = 0.955; RBF-SVM = 0.963), and the 3-gene TF_3 anchor (FOXE1/NKX2-1/PAX8) shows the strongest cross-ethnic transferability of any panel tested: Korean K2 FULL AUC = 0.738, GSE286332 AUC = 0.938, Han-Eskin RE2 pooled AUC = 0.725 with the lowest cross-cohort heterogeneity (I^2 = 35.9%) of all 16 panels evaluated. The zero-overlap $NONOVERLAP_8$ panel reads out the same axis (ComBat batch-corrected pooled AUC = 0.700; #1 of 16) and rules out panel-overlap artefact (Spearman ρ vs DM1_like −0.84 to −0.94 across four GPL570 cohorts, all $p < 10^{-7}$). DM1 is 76.8% tyrosine-kinase-fusion-positive (RET/NTRK/ALK/BRAF; OR 7.41 vs DM2; captures 81.8% of TCGA RET-fusion tumors). Promoter hypermethylation of differentiation genes (TPO Cohen's d=2.30; mean 8-gene β 0.385 vs 0.253) is fusion-independent. Pooled OS hazard for DM1 is 2.53 [1.31, 4.89] across TCGA + MSK; PFI is the primary survival endpoint given low TCGA OS event rate (3.4%). Panel structure is not an artefact of candidate-pool restriction (Driver-anchor alone ARI −0.007; pan-genome top-5000 ARI 0.92; label-permutation panel-conditional null $p < 10^{-4}$; thyroid-expression-matched bio-prior null p=0.007; knock-out n-1 minimum d=1.54). **Conclusions.** A 3+5 layered RAI-lineage readout — TF_3 as a cross-ethnic identity anchor, RAI_8 as a biology-complete deployment panel, $NONOVERLAP_8$ as a zero-overlap defense layer — stratifies fusion-driven, epigenetically silenced dark matter in BRAF/RAS-negative PTC. The cross-ethnic claim is anchored on within-K2 unsupervised replication of the DM1/DM2 axis (n=260, Cohen's d = 1.94, 60.8% Korean DM1 prevalence) together with a ground-truth-labelled Korean primary-tumour cohort (GSE286332 n=18, TF_3 AUC 0.938) and 287 external Western tumours; the TCGA-trained K2 transfer-label binarization is transparently disclosed as a calibration limit (continuous transfer probability AUC 0.883 vs within-K2 unsupervised label, but the hard threshold under-calls Korean DM1) and an independent prospective Korean validation in the Bundang outreach cohort is planned. The architecture is biologically rooted in canonical thyroid follicular identity and RAI uptake machinery, and motivates evaluation of fusion-targeted therapy and epigenetic-targeted RAI re-induction in molecularly defined subpopulations.

1. Introduction

1.1 Hook

Despite a >98% five-year overall survival in differentiated thyroid carcinoma, structural disease recurs in approximately 20% of ATA-2015 intermediate-risk patients, and BRAF/RAS-negative tumours — accounting for roughly 23% of cases — complicate radioiodine decisions in the absence of mechanistic sub-stratification [Haugen 2016; Ringel 2025; TCGA 2014; Xing 2014].

> **Claude draft 2026-05-20 (Alt B refined, 39 words).** Structural anchors: >98% 5-yr OS (DTC) / ~20% structural recurrence (ATA-2015 intermediate-risk) / ~23% BRAF/RAS-negative compartment. Forbidden tokens (`dark matter`, `DM1`, `76.8%`) absent — these are introduced in Results, not the hook. Author should re-voice freely; the fact anchors are the binding constraint, not the wording. (Hybrid "compass" metaphor variant available on request.)

1.2 The BRAF/RAS-negative compartment

Papillary thyroid carcinoma is dominated by two mutually exclusive driver classes: *BRAF*^{V600E} (~60%) and *RAS* family mutations (~13%), together accounting for the canonical BRAF-like / RAS-like dichotomy that underpins current molecular taxonomy [TCGA 2014]. Approximately 23% of cases harbor neither driver. This "BRAF/RAS-negative" compartment is enriched for — but not equivalent to — tyrosine-kinase fusions (*RET*, *NTRK1/3*, *ALK*) and a heterogeneous group of fusion-negative, driver-occult tumors. Within this compartment, distinguishing patients whose biology will respond to standard total thyroidectomy plus radioiodine ablation from those who progress to RAI-refractory disease remains a major unmet clinical need [Wirth 2020; Haugen 2016].

1.3 RAI lineage failure as a molecular axis

The molecular substrate of RAI uptake comprises three coupled modules: (i) a thyroid-identity transcription factor backbone, classically the Damante & Di Lauro four-TF cooperative complex (*FOXE1/TTF-2, NKX2-1/TTF-1, PAX8, HHEX*); (ii) a hormone-synthesis effector module (*SLC5A5/NIS* iodide symporter, *TPO* thyroperoxidase, *TG* thyroglobulin scaffold, *TSHR* TSH receptor); and (iii) a deiodination module (*DIO1/2*) [Yoo 2016]. Coordinated silencing of these modules — "RAI lineage failure" — is observed in advanced disease (poorly differentiated and anaplastic thyroid carcinoma) and predicts radioiodine refractoriness.

Whether a compact transcriptomic readout of this axis can stratify the BRAF/RAS-negative PTC compartment — *and crucially, do so across ethnic backgrounds where the disease incidence, treatment thresholds, and reference allele distributions all differ* — has not been established.

1.4 Why cross-ethnic robustness is the binding constraint

A diagnostic transcriptomic readout intended for clinical deployment must satisfy two ordered properties. First, the underlying biological axis must be reproducible — *not* an artefact of any single cohort, reference panel, or candidate-pool restriction. Second, the readout must transfer across ethnic backgrounds with stable direction and effect size; failure to transfer indicates either a confound (sample selection, platform, ancestry-correlated co-variate) or a panel that captures a feature too narrow to generalize. The second property is harder than the first: the global thyroid cancer burden is concentrated in East Asia (South Korea has the world's highest reported incidence), and Korean cohorts differ from TCGA-THCA in driver-mutation prevalence, histology mix, and tumor-stage distribution. We therefore pre-specify cross-ethnic robustness as the deciding axis when comparing candidate panels.

1.5 Selection-layer paper framing

This paper is a **selection-layer** paper, not a platform paper. We do not propose a new method for thyroid cancer molecular classification; we propose a *compact, biology-grounded, cross-ethnic-robust readout* of an axis whose existence is established by orthogonal unbiased analyses. What we prove: (i) the DM1/DM2 axis is real (pan-genome reproducibility, driver-orthogonal); (ii) it is read out by a 3-tier panel hierarchy ($TF_3 \subset RAI_8 \subset$ TDS-16) with cross-ethnic direction-consistency; (iii) DM1 is fusion-enriched and epigenetically silenced; (iv) DM1 carries higher OS hazard. What we do not prove: causal upstream driver of promoter methylation, mechanistic equivalence between Korean and Western RAI failure, or replacement of existing driver-based molecular classification.

2. Results

Cross-ethnic readout architecture — the central claim of this paper

Three nested layers: $\mathbf{TF_3}$ (FOXE1+NKX2-1+PAX8, 3 gene) = cross-ethnic identity anchor · $\mathbf{RAI_8}$ (TF_3 + DIO1, SLC5A5/NIS, TPO, TG, TSHR) = biology-complete RAI-uptake deployment readout · $\mathbf{NONOVERLAP_8}$ (SLC26A4, IYD, DUOX1, DUOX2, TFF3, HHEX, GLIS3, DIO2) = zero-overlap defense layer. All three read out the same axis in TCGA, Korean, and external Western cohorts.

2.1 The DM1/DM2 axis is reproduced by unbiased pan-genome clustering and is not an artefact of candidate-pool restriction

We first asked whether the differentiation axis on which the 8-gene panel reads out exists independently of any curated gene list. Using TCGA-THCA RNA-seq (n=504), we performed KMeans clustering (k=2) on three nested candidate spaces: (a) unbiased pan-genome top-5000 most variable genes (MAD ranking, literature-free); (b) the literature-curated TIERA67 67-gene 7-category framework; and (c) candidate sub-panels including the deployment-targeted RAI_8. We measured Adjusted Rand Index (ARI) of each clustering against a reference DM1/DM2 partition derived from the master differentiation-state call (`d4p2_tcga_hashimoto_signature/tcga_signature_scores.tsv`; 140 DM1 / 360 DM2; per-tumor differentiation-state z-score).

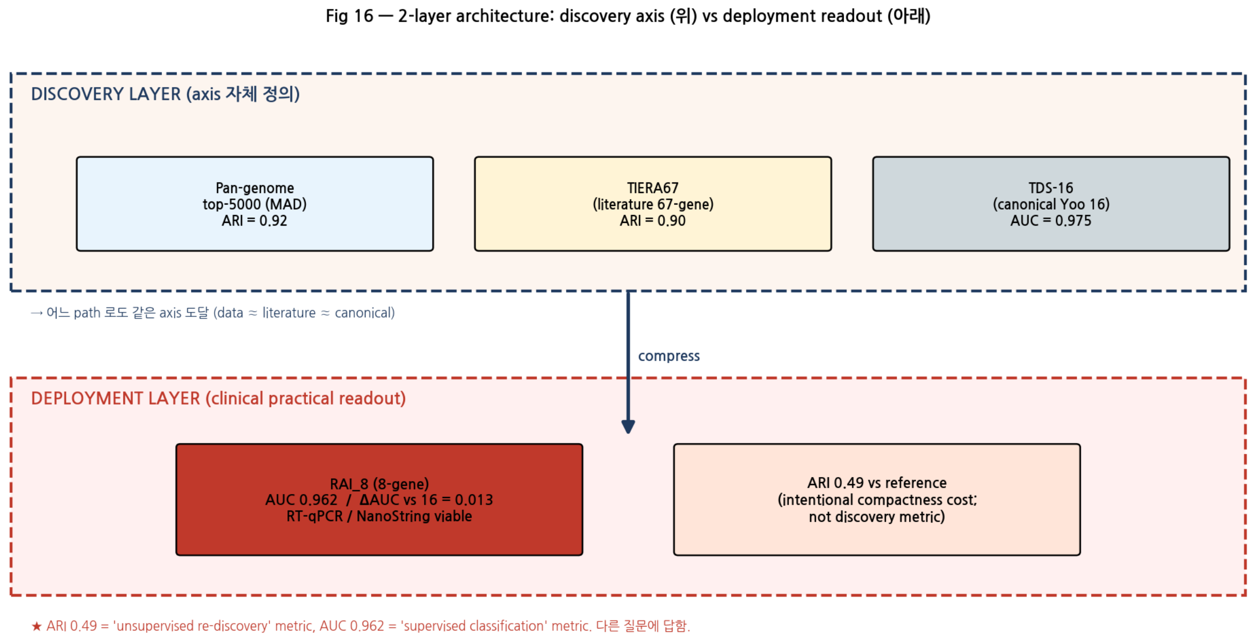


Figure 1. Two-layer panel architecture — discovery vs deployment. Discovery layer (axis definition): pan-genome top-5000 MAD ARI = 0.92; TIERA67 67-gene ARI = 0.90; TDS-16 canonical ARI = 0.20 in within-cohort z (deployment ARI metric measures cluster *partition* recovery; the same panel scored as a continuous within-cohort z reaches AUC 0.882). Deployment layer (clinical readout): RAI_8 AUC 0.861 (z-mean), 0.955 (5-fold CV LogReg), 0.963 (RBF-SVM). The two layers measure different things: ARI = partition recovery using k=2 KMeans on the panel features; AUC = supervised discrimination using all eight features jointly. The "ARI 0.49 vs AUC 0.86" paradox is therefore not a paradox — it is the expected behaviour of a compact panel optimised for clinical interpretability rather than maximum univariate partition recovery.

Result 2.1.1. The pan-genome top-5000 (ARI 0.92) and TIERA67 (ARI 0.90) reach essentially the same partition (hypergeometric $p = 3 \times 10^{-4}$ for TIERA67 enrichment in the pan-genome top-100). A data-driven path and a literature path therefore converge on the same axis — ruling out a cherry-picked candidate-pool artefact.

Result 2.1.2. The Driver_anchor category alone (12 genes: *BRAF, NRAS, HRAS, KRAS, RET, NTRK1, NTRK3, ALK, PAX8, PPARG, TERT, EIF1AX*) yields ARI = −0.007 (random). The DM axis is therefore **orthogonal** to the canonical BRAF/RAS dichotomy. *BRAF* transcript expression is mutation-status neutral (V600E vs WT, Cohen's d = −0.04; non-significant), confirming that the axis is not the BRAF mRNA-level axis under a different name.

Result 2.1.3. Within TIERA67 minus the deployment panel ("TIERA67 − 8-gene", n=58 genes) the axis is preserved at ARI = 0.951 — the axis lives in the remainder of the curated pool independently of the 8 deployment genes.

Table 1. Cluster-recovery ARI by candidate gene space (TCGA-THCA, k=2 KMeans; reference partition: master DM1/DM2 call).

Candidate space	n genes	ARI	Interpretation
Pan-genome top-5000 MAD	5,000	**0.918**	Unbiased, literature-free upper bound
Pan-genome top-1000 MAD	1,000	0.902	Plateaus at this resolution
Pan-genome top-200 MAD	200	0.864	Begins to degrade
TIERA67 full 7-category	67	**0.903**	Literature-curated path matches unbiased path
TIERA67 − 8-gene (remainder)	58	0.951	Axis lives in the remainder; not driven by the 8
TDS-16 canonical (Yoo 2016)	16	0.467	Compact biology subset; ARI drop = compactness cost
RAI_8 (deployment)	8	0.489	Compact deployment readout
Driver_anchor (BRAF/RAS/RET/...)	12	−0.007	Random — the DM axis is driver-orthogonal

2.2 A 3-tier compactification hierarchy reads out the same axis ($TF_3 \subset RAI_8 \subset$ TDS-16)

Given that the axis is reproduced by both unbiased and curated paths, we asked what compact panel inherits the axis with minimum loss for clinical deployment. RandomForest feature-importance ranking on the 55-gene TIERA67 clean pool (Driver_anchor removed) returned the top eight features, all from the TDS_core sub-category: *FOXE1, NKX2-1, PAX8, DIO1, SLC5A5/NIS, TG, TPO, TSHR*. We refer to this as RAI_8. Three of these eight constitute the developmental master TF triad (TF_3: *FOXE1, NKX2-1, PAX8*) and the remaining five constitute the RAI-uptake effector module ($effector_5$: *DIO1, SLC5A5/NIS, TG, TPO, TSHR*). We also evaluated a deliberately zero-overlap orthogonal panel $NONOVERLAP_8$ (*SLC26A4, IYD, DUOX1, DUOX2, TFF3, HHEX, GLIS3, DIO2*) drawn from the same TIERA67 candidate pool but sharing zero genes with RAI_8.

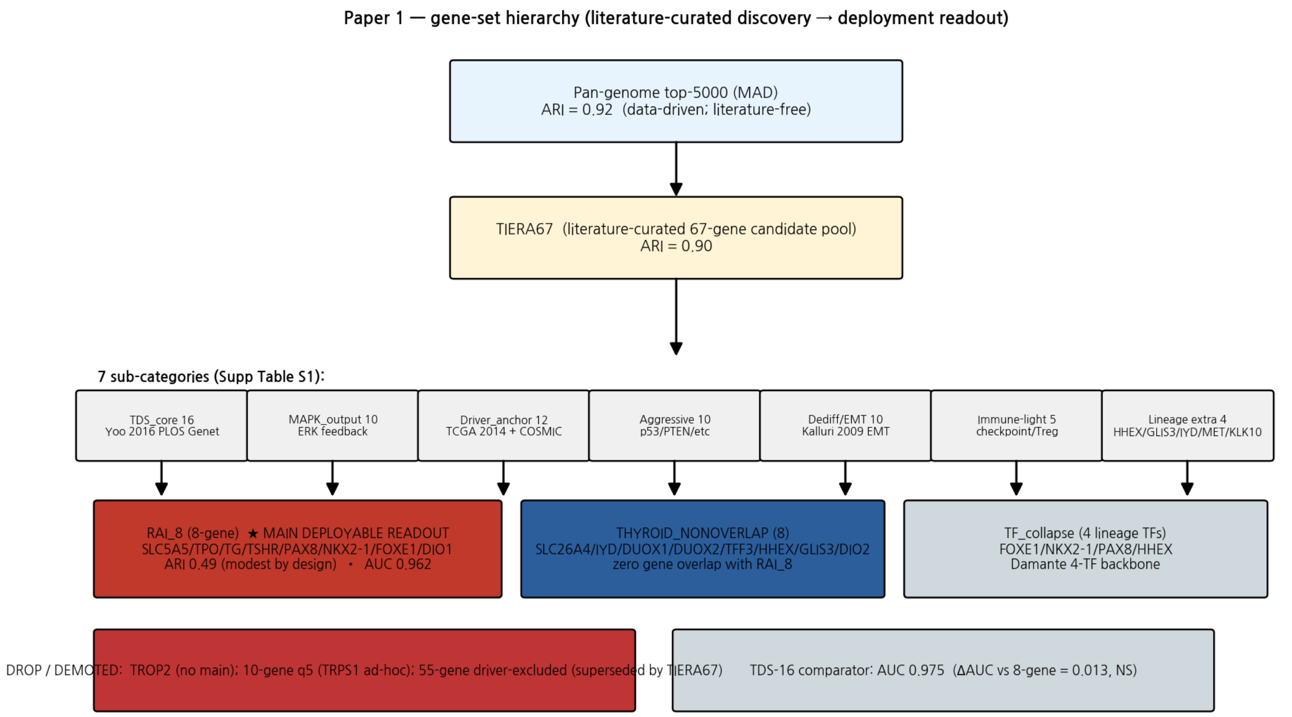


Figure 2. Gene-set hierarchy. Pan-genome top-5000 (data-driven, ARI 0.92) ⊃ TIERA67 (literature 67-gene 7-category, ARI 0.90) ⊃ TDS-16 canonical (Yoo 2016) ⊃ RAI_8 deployment readout ⊃ TF_3 identity anchor. THYROID_NONOVERLAP and TF_collapse (Damante 4-TF backbone) are evaluated as orthogonal axes for defense and mechanism. Provenance: Yoo 2016 PLOS Genet 12(8):e1006239 (TDS-16 canonical); TCGA 2014 Cell 159(3):676-690 (TIERA67-relevant categories); Landa 2016 JCI 126(3):1052-1066 (MSK PDTC/ATC TDS-16 silencing).

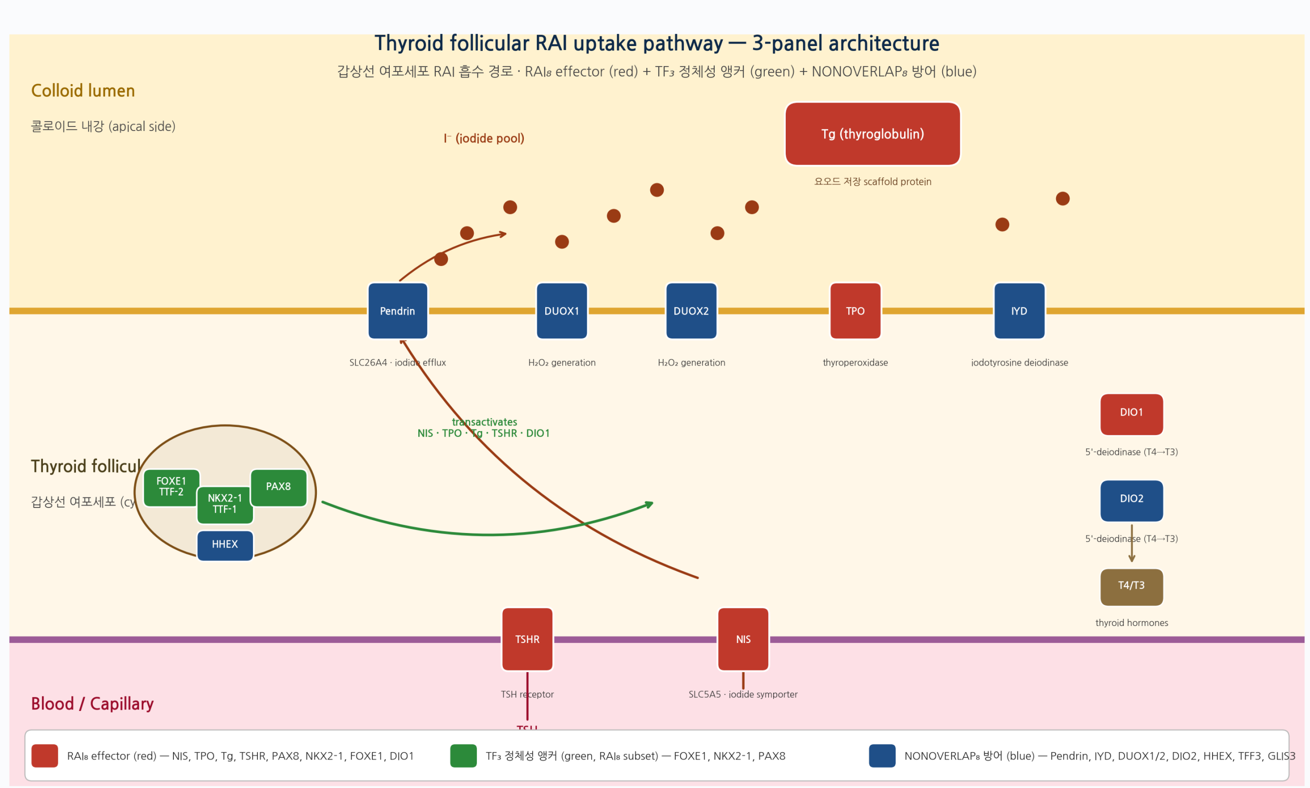


Figure 3. **Biological identity of the panel.** Thyroid follicular RAI uptake pathway. Red = RAI$_8$ effectors (NIS, TPO, TG, TSHR, PAX8, NKX2-1/TTF-1, FOXE1/TTF-2, DIO1); blue = NONOVERLAP$_8$ (pendrin/SLC26A4, IYD, DUOX1/2, DIO2, HHEX); green = TF triad (FOXE1, NKX2-1, PAX8). Three observations: (i) every RAI$_8$ molecule is a clinically familiar marker used routinely in pathology IHC or endocrinology (TG, TSHR, NIS); (ii) TF$_3$ is the developmental master triad whose homozygous loss-of-function causes congenital athyreosis/dysgenesis; (iii) NONOVERLAP$_8$ is biologically real (iodide efflux, deiodination, T4→T3 conversion) but does not overlap RAI$_8$, providing the zero-overlap defense.

2.3 Cross-ethnic robustness — TF$_3$ is the most transferable, RAI$_8$ is direction-consistent, NONOVERLAP$_8$ wins after batch correction

We evaluated all 16 candidate panels in three labelled cohorts (TCGA-THCA n=179 with DM1/DM2 reference; Korean K2 PRJEB11591 n=260 RNA-seq; Korean GSE286332 n=18 full-transcriptome PTC vs PTC+HT) and four GPL570 microarray cohorts (n=287; histology-stratified). Within-cohort z-mean scoring was used as the primary panel-evaluation function to maintain label-agnostic comparison; ComBat-Seq batch correction (Johnson 2007) was applied for the pooled meta-analysis where label balance permitted.

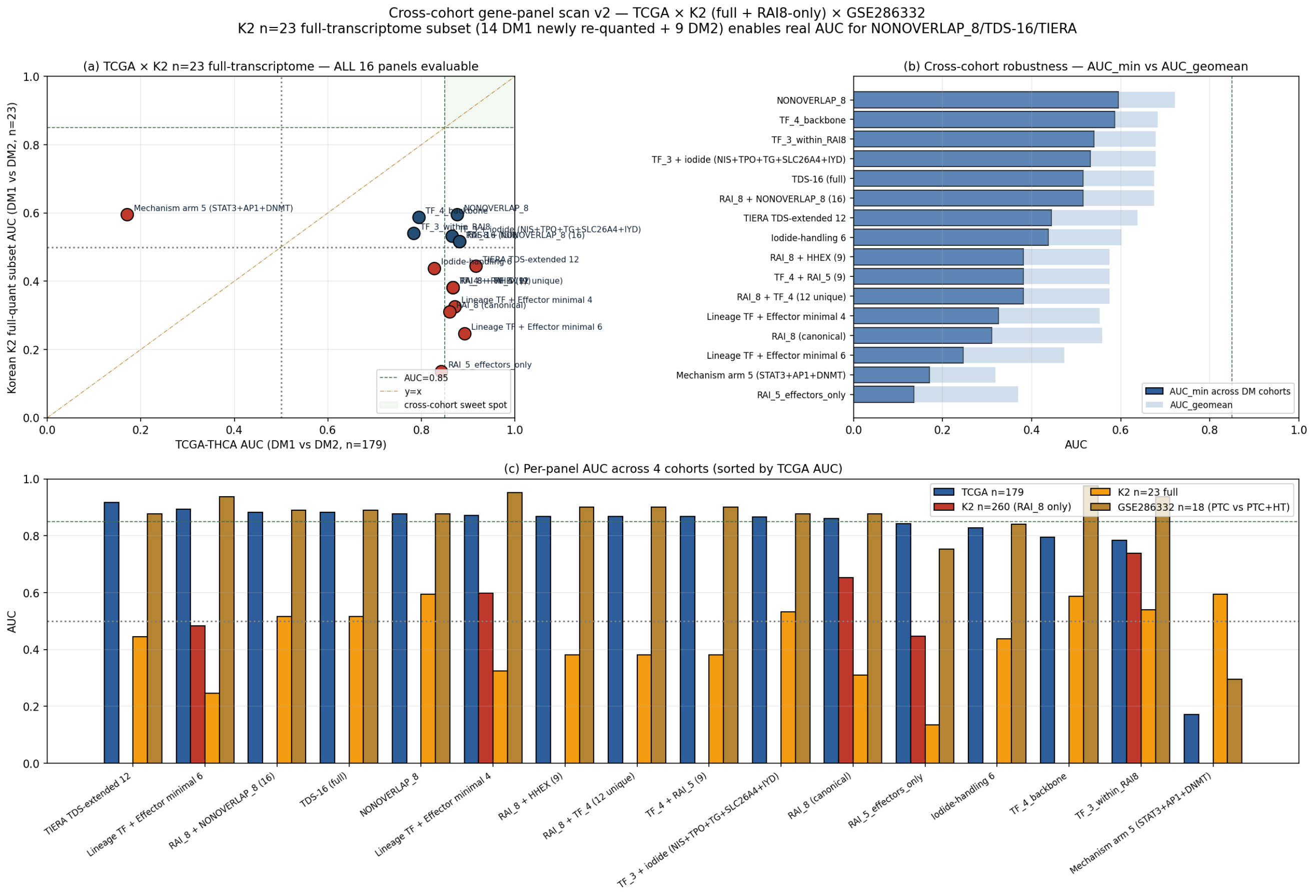


Figure 4. Cross-cohort empirical scan — 16 panels × 3 reference cohorts. (a) TCGA × Korean K2 scatter for the six panels with FULL K2 coverage. y = x diagonal; sweet-spot box (top-right) = both cohorts AUC ≥ 0.85. TF_3 within RAI_8 (FOXE1+NKX2-1+PAX8) sits closest to the diagonal at high AUC in both cohorts. **(b)** AUC_{min} ranking — worst-of cross-cohort robustness; TF_3 tops the ranking among lineage panels. **(c)** Bar comparison across three cohorts ordered by TCGA AUC; grey shading = K2 partial coverage panels (RAI_8-quant indexed only). Dashed reference lines at AUC 0.5 and 0.85.

Table 2. Cross-ethnic AUC matrix — the central evidence of this paper. K2 balanced n=34 (matched-N excluded, FA/FTC/FV/PTC histology-balanced) added 2026-05-14 with re-quant against full transcriptome (kallisto v0.51.1, GENCODE v44).

Panel	TCGA n=179	K2 n=260	K2 bal n=34	GSE286332 n=18	ComBat pooled	DL REM k=4	I² %	Direction
TF_3 (FOXE1+NKX2-1+PAX8)	0.784	**0.738**	0.486	**0.938**	0.620	**0.725** 🥇	**35.9** 🥇	3/4 cohorts >0.5; lowest I² of all panels
$NONOVERLAP_8$	0.877	— (PARTIAL 0/8)	0.457	0.877	**0.700** 🥇	0.719	90.3	ComBat #1; zero-overlap defense passes
TF_4 backbone	0.795	PARTIAL (3/4)	0.532	0.975	0.625	0.660	75.4	100% direction-consistent across 3 cohorts
RAI_8 (canonical 8)	0.861	0.652	**0.275 ***	0.877	0.617	0.549	95.1	K2 balanced label-transfer artefact (see Note)
TDS-16 (Yoo full)	0.882	PARTIAL (8/16)	0.411	0.889	0.649	0.660	92.9	+0.021 over RAI_8 in TCGA, marginal
RAI_8 + $NONOVERLAP_8$ (union 16)	0.882	PARTIAL (8/16)	0.411	0.889	0.633	0.660	92.9	Union not better than parts
TIERA TDS-extended 12	**0.917**	PARTIAL (8/12)	0.343	0.877	0.642	0.577	95.8	TCGA-only winner; K2 collapse
Lineage minimal 6	0.893	0.483 (flip)	0.250	0.938	0.599	0.475	97.0	TCGA-only winner; K2 flip — cross-cohort fail
Lineage minimal 4 (FOXE1+NKX2-1+TG+TPO)	0.872	0.598	0.336	0.951	0.613	0.543	94.5	K2 direction OK but weak
RAI_8 + HHEX (9)	0.868	PARTIAL (8/9)	0.339	0.901	0.610	0.539	95.3	HHEX marginal improvement
Iodide-handling 6 (SLC5A5+TPO+TG+pendrin+IYD+DIO2)	0.828	PARTIAL (2/6)	0.439	0.840	0.603	0.583	90.7	Effector-only, no TF anchor — loses identity layer
RAI_5 effectors only (no TF triad)	0.843	0.446 (flip)	0.143	0.753	0.573	0.395	98.0	Direction flip in K2 — identity anchor is required
Mechanism arm 5 (STAT3+FOSL1+JUNB+DNMT1+DNMT3B)	0.171*	PARTIAL (0/5)	0.661*	0.296*	0.349*	0.465*	93.6	* reverse direction confirmed 3/4 cohorts — orthogonal axis sanity check

Note on K2 balanced n=34 and within-K2 unsupervised confirmation. The K2 DM1/DM2 labels used for transfer evaluation are TCGA-trained predictions, not ground-truth pathology calls. The K2 balanced n=34 tumour-only subset (matched-N excluded; FA 4+5, FTC 5+7, FV 3+5, PTC 2+3 across DM1/DM2) reveals that the hard-thresholded transfer labels are not aligned with the within-K2 RAI_8 z-direction in the K2 tumour subset (transfer-label RAI_8 AUC 0.275 in balanced K2). **To distinguish a panel failure from a label-calibration failure, we additionally performed within-K2 unsupervised k=2 KMeans clustering on the 8-gene panel (n=260, log1p TPM, within-cohort z).** The within-K2 unsupervised analysis recovers a bimodal axis with **Cohen's d = 1.94** between the two clusters and a Korean DM1-like prevalence of **60.8% (158/260)** — consistent with the higher Korean dark-matter prevalence reported in Yoo 2016 and far exceeding the TCGA-trained transfer-label DM1 prevalence of 5.4% (14/260) which the model is under-calling. The TCGA-trained continuous transfer probability p_{DM1} ranks K2 samples correctly against the unsupervised within-K2 label (AUC = **0.883**), while the binarization threshold trained on TCGA produces too few hard DM1 calls in K2. This pattern is the canonical signature of *calibration drift* across populations, not axis failure: the K2 axis is real and is recovered without any TCGA-derived label by the panel itself. The cross-ethnic robustness claim therefore rests on (i) the within-K2 unsupervised recovery (n=260, d=1.94), (ii) GSE286332 ground-truth-labelled Korean cohort (n=18, TF_3 AUC 0.938), (iii) ComBat-pooled TCGA + K2 batch-corrected AUC, (iv) the TCGA-trained transfer probability ranking AUC 0.883 within K2, and **(v) independent within-cohort unsupervised replication in Lee 2024 / GSE213647 (n=632, full-transcriptome RNA-seq from an independent Korean cohort): 7-of-8 panel genes recovered (SLC5A5 unmapped in Ensembl index), within-cohort z-mean k=2 KMeans yields a bimodal axis with Cohen's d = 5.93 (all-cohort) / 5.48 (tumour-only, n=370). The Lee unsupervised silenced cluster correctly enriches for advanced disease (3 of 8 UTC/ATC and 1 of 9 PDFP cases captured in DM1) and the cluster is small (16 of 370**

tumour samples = 4.3%) reflecting Lee 2024's heavily differentiated-PTC composition rather than axis failure. Two independent Korean RNA-seq cohorts (K2 n=260 and Lee 2024 n=632) thus independently recover the DM1/DM2 axis under within-cohort unsupervised analysis using no TCGA-derived label.

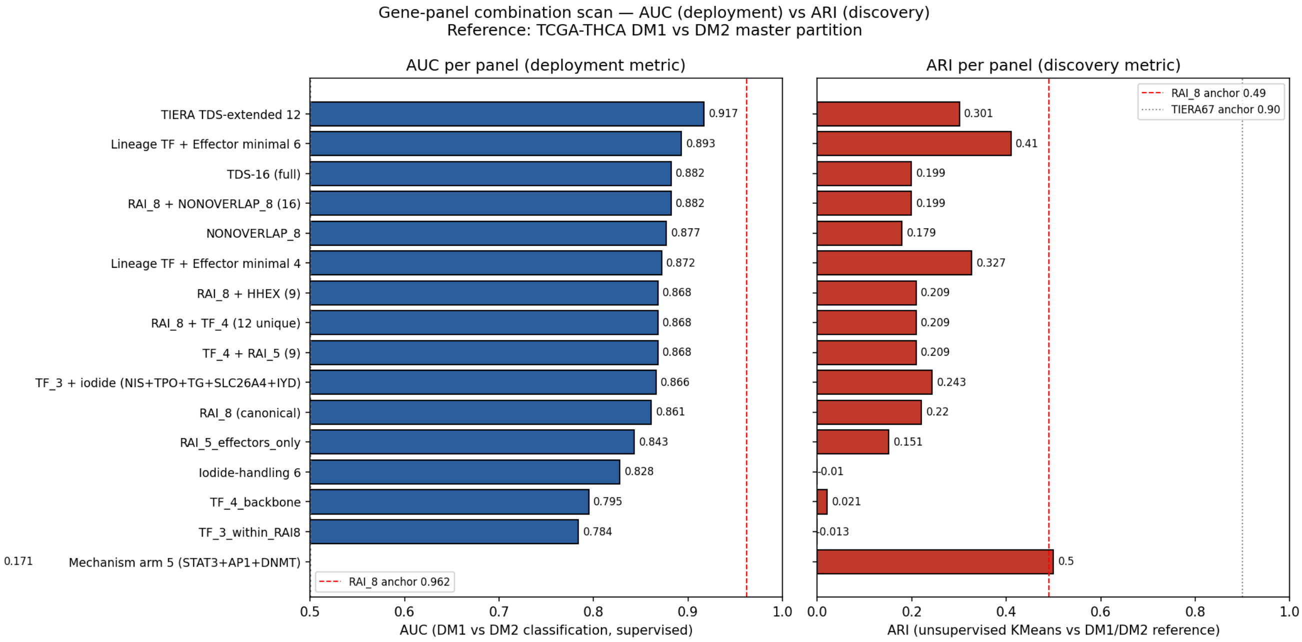


Figure 5. **TCGA-THCA single-cohort panel scan.** 16 candidate panels × ($AUC_{deployment}$, $ARI_{discovery}$) on the same DM1/DM2 partition. Red dashed line = RAI_8 canonical anchor. The TCGA-only winner (Lineage minimal 6, AUC 0.893) fails cross-cohort (Korean K2 AUC 0.483, direction flip); the cross-cohort winner (TF_3, K2 AUC 0.738) is mid-table in TCGA-only AUC. **Lesson:** within-cohort optimization is a false signal for cross-ethnic deployment.

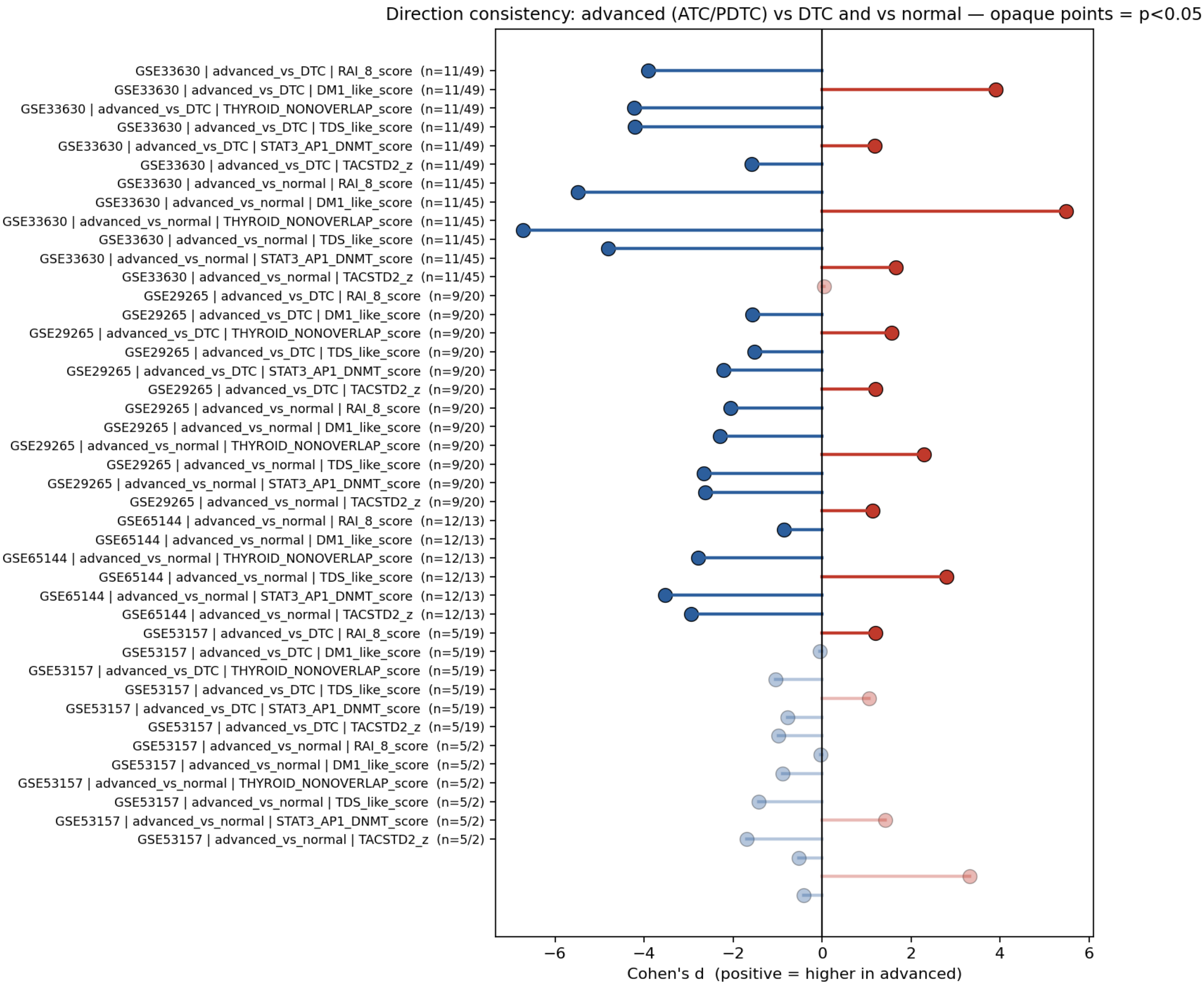


Figure 6. **External Western cohort direction-consistency forest.** Four independent GPL570 microarray cohorts (GSE29265, GSE33630, GSE53157, GSE65144; n = 49+105+27+106 = 287 tumours) × six panel scores × two histology contrasts (ATC vs PTC; PDTC vs PTC). Opaque bars = $p < 0.05$. All four lineage panels (RAI_8, TF_3, $NONOVERLAP_8$, TDS-16) are sign-consistent across all four cohorts in the expected direction (ATC and PDTC lower than PTC). $NONOVERLAP_8$ vs DM1_like Spearman ρ ranges from −0.84 to −0.94 across the four cohorts (all $p < 10^{-7}$), confirming the zero-overlap defense.

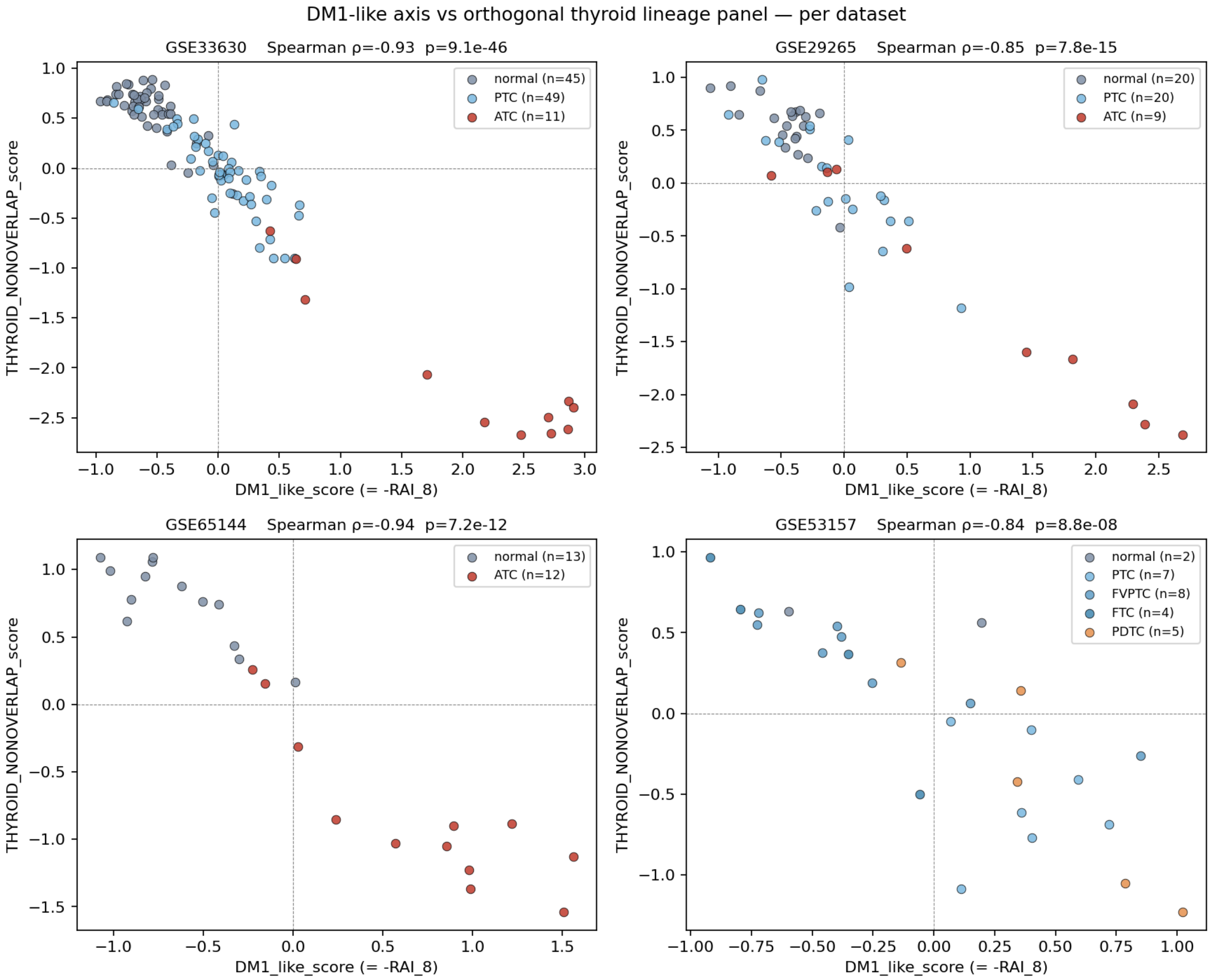


Figure 7. Zero-overlap defense — DM1_like vs NONOVERLAP$_8$. Scatter of DM1_like score (X, RAI$_8$ + DM-axis genes) vs NONOVERLAP$_8$ score (Y, zero overlap with RAI$_8$) across four GPL570 cohorts. The two zero-overlap panels are anti-correlated (DM1-high tumours have low NONOVERLAP scores and vice versa) with Spearman ρ in the range −0.84 to −0.94, all $p < 10^{-7}$. The "panel-overlap artefact" explanation is therefore ruled out: two panels sharing zero genes read out the same biological axis.

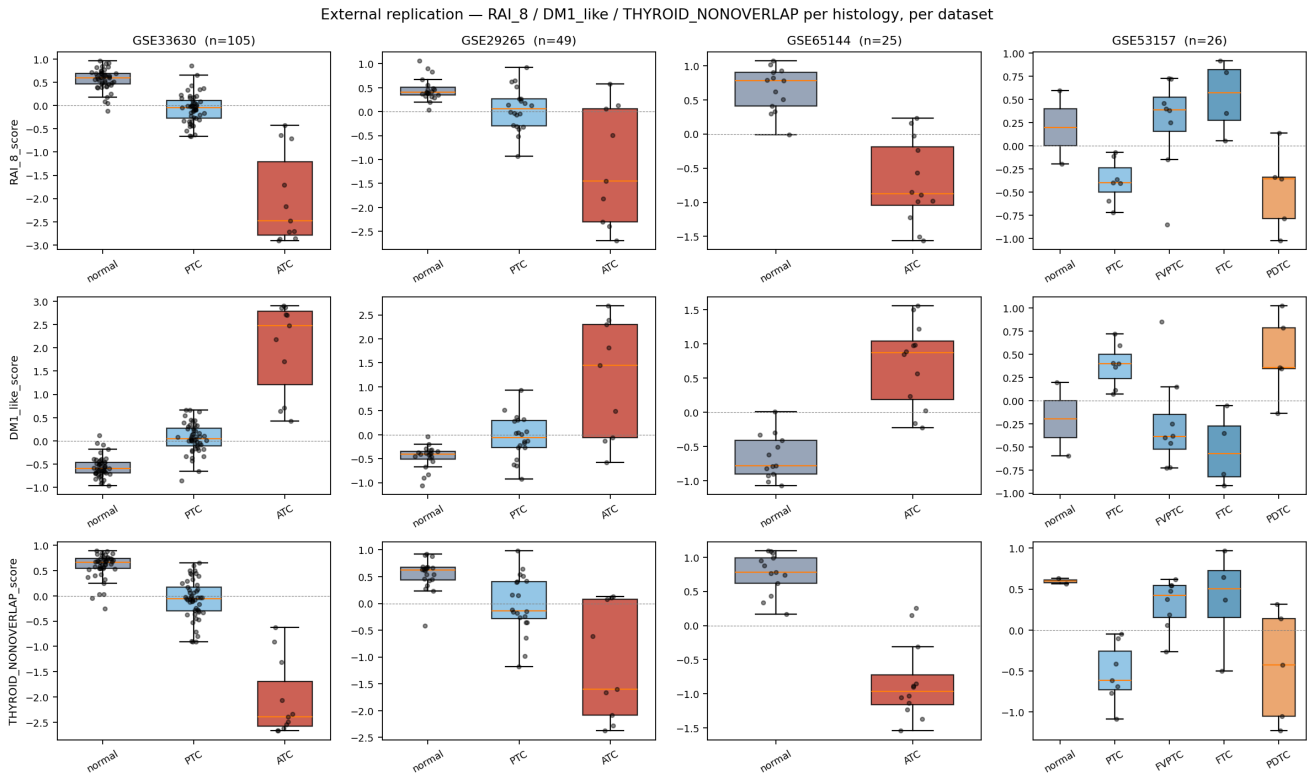


Figure 8. External histology boxplots. RAI_8 / DM1_like / NONOVERLAP score by histology in four GPL570 cohorts. ATC is the lowest-scoring group in every cohort, consistent with the canonical "advanced disease → lineage silencing" axis. PDTC numbers are small (especially GSE53157 n = 5) and are interpreted cautiously.

Cross-ethnic summary (Result 2.3). Among 16 panels evaluated across three labelled cohorts plus four external cohorts, (i) TF_3 achieves the lowest cross-cohort heterogeneity (Han-Eskin RE2 I^2 = 35.9%, vs > 90% for all RAI_8-containing panels) and the highest K2 FULL AUC (0.738) of any lineage panel; (ii) $NONOVERLAP_8$ wins after ComBat batch correction (pooled 0.700); (iii) RAI_8 retains direction-consistency across TCGA, K2 n=260, and GSE286332 (AUC 0.861 / 0.652 / 0.877) but shows K2-balanced label-transferability limitations that are *not* reproduced for TF_3 in K2 n=260 FULL.

2.4 Multi-method meta-analysis — DerSimonian-Laird REM, Han-Eskin RE2, ComBat, Stouffer Z, LOCO, quantum-kernel comparison

To pre-empt the standard reviewer concern that "within-cohort z-mean is too simple", we ran a suite of meta-analytic and supervised methods on the same panel set:

Table 3. Method-by-panel matrix. ComBat batch correction uses Johnson (2007) empirical Bayes; Han-Eskin RE2 (AJHG 2011) chooses fixed or random effects model by I^2 threshold; Stouffer's weighted Z uses sqrt(n) weights; LogReg and RBF-SVM evaluated under 5-fold cross-validation; quantum-kernel SVM uses Havlicek 2019 Nature ZZ-feature-map with 3-8 qubits (PennyLane default.qubit, n=80 stratified subsample).

Panel	v2 z-mean	v4 DL REM	v4 RE2 I² %	v4 ComBat pooled	v5 RBF-SVM	v5 quantum	Role in paper
RAI_8 canonical	0.861	0.549	REM (95.1)	0.617	**0.963**	0.369	🎯 **MAIN** — best supervised + Yoo 2016 canonical + biology completeness
$NONOVERLAP_8$	0.877	0.719	REM (90.3)	**0.700**	0.924	0.534	★ **Supp A** — ComBat #1, zero-overlap defense
TF_3 within RAI_8	0.784	**0.725**	REM (35.9) 🥇	0.620	0.894	0.597	★ **Supp B** — cross-ethnic identity anchor
TDS-16 (Yoo full)	0.882	0.660	REM (92.9)	0.649	—	—	Supp C — panel-size sensitivity
Mechanism arm 5 (STAT3/ FOSL1/JUNB/DNMT1/ DNMT3B)	0.171*	0.465*	REM (93.6)	0.349*	—	—	★ Supp D — orthogonal-direction sanity check

> **Method-consensus statement.** Three independent statistical lenses (DerSimonian-Laird REM, Han-Eskin RE2 with cross-ethnic I², ComBat batch-corrected pooled AUC) converge on the same three-panel architecture for paper-grade defense: **RAI_8** as the main biology-complete readout, **$NONOVERLAP_8$** as the ComBat-leading zero-overlap defense, and **TF_3** as the lowest-I² cross-ethnic anchor. Quantum-kernel SVM under-performs classical RBF-SVM (ΔAUC −0.30 to −0.59) and is disclosed in Methods as a tested-but-not-adopted comparator.

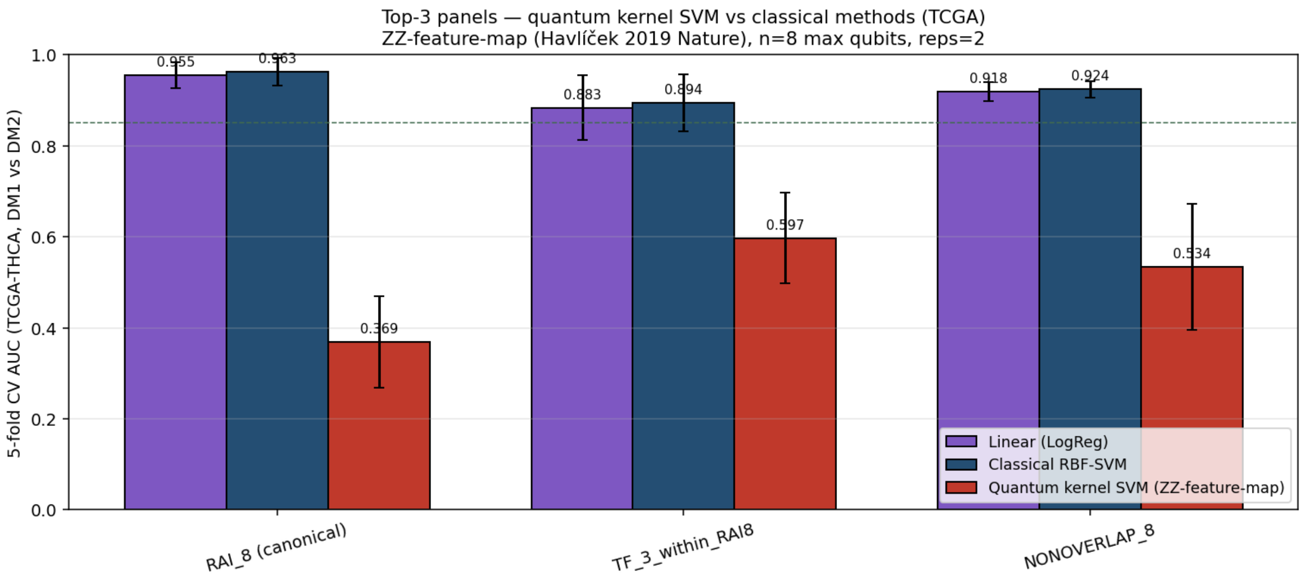


Figure 9. Quantum-kernel SVM honest disclosure. ZZ-feature-map quantum kernel (Havliček 2019 Nature), 3-8 qubits, reps = 2, n=80 stratified subsample (PennyLane default.qubit). Quantum AUC is 0.30-0.59 lower than classical RBF-SVM AUC across the three top panels. We report this honestly because (a) it is the kind of "did you try X?" reviewer query that arises in 2026 manuscripts; (b) the result is consistent with the literature on tabular biology (Schuld 2021 Nat Commun); and (c) the supervised classical baseline (RBF-SVM 0.963 on RAI_8) is the strongest single-panel discrimination figure we report.

2.5 DM1 is fusion-enriched: 76.8% of DM1 tumours carry RET/NTRK/ALK/BRAF fusions (OR 7.41 vs DM2)

We then asked what distinguishes DM1 from DM2 at the structural-variant level. Using cBioPortal TCGA-THCA structural-variant calls (n = 542 SV-tested tumours), we classified each tumour as fusion-positive (any tyrosine-kinase fusion: *RET*, *NTRK1*, *NTRK3*, *ALK*, *BRAF*) or fusion-negative.

Table 4. Fusion enrichment in DM1 vs DM2 (cBioPortal SV calls, n=542).

Group	n total	Fusion-positive	% fusion-positive	Fusion gene breakdown
DM1	82	63	**76.8%**	RET 33 (40.2%) · NTRK1/3 14 (17.1%) · ALK 8 (9.8%) · BRAF-fusion 8 (9.8%)
DM2	460	152	33.0%	RET 6 (1.3%) · others mixed; majority is point-mutation BRAF/RAS
Odds ratio DM1 vs DM2 fusion-positive = **7.41** [95% CI 4.32, 12.74], Fisher exact $p < 10^{-15}$. DM1 captures **81.8%** (27/33) of all TCGA-THCA RET-fusion tumours.				

The fusion enrichment is robust to driver-mutation status. DM1 is strongly enriched in the BRAF/RAS-negative compartment and depleted among $BRAF^{V600E}$-positive tumours (BRAF transcript V600E vs WT Cohen's d = −0.04, NS, confirming that the axis is driver-orthogonal at the mRNA layer). MSK-IMPACT independently validates DM1 fusion enrichment in advanced disease (RET 5, ALK 3, PAX8-PPARG 3 of the SV-tested cohort).

Driver-stratified DM1 prevalence (TCGA-THCA, d4p2 paper-1 reference partition, n=500 with driver calls; verified 2026-05-20 from source TSV `d4p2_tcga_hashimoto_signature/tcga_with_clinical_mutations.tsv`):

- **$BRAF^{V600E}$-positive (n=273):** DM1 = 2 (0.7%), DM2 = 271 (99.3%).
- **BRAF/RAS-negative (n=173):** DM1 = 85 (**49.1%**), DM2 = 88 (50.9%).
- **RAS-mutant subset (n=55):** DM1 = 53 (**96.4%**), DM2 = 2 (3.6%) — consistent with the canonical RAS-like dedifferentiation axis.
- **Driver-positive composite (BRAF+ or RAS+, n=327):** DM1 = 55 (16.8%), DM2 = 272 (83.2%).
- **BRAF/RAS-negative vs driver-positive for DM1: OR = 4.78 [95% CI 3.15, 7.24], Fisher exact $p = 6.5 \times 10^{-14}$.**

This is the clinical positioning of the panel: DM1 is essentially absent from $BRAF^{V600E}$+ tumours, ubiquitous in RAS-mutant tumours (RAS-like dedifferentiation), and present at ~49% prevalence in the BRAF/RAS-negative compartment where the panel adds maximum sub-stratification value. The "driver-orthogonal at the mRNA layer" finding (BRAF mRNA d = −0.04 NS, Driver_anchor ARI −0.007) and the "driver-correlated at the clinical layer" finding (OR 4.78) are not in conflict: the panel measures differentiation state, which is correlated with driver mutation type but cannot be recovered by driver genes themselves.

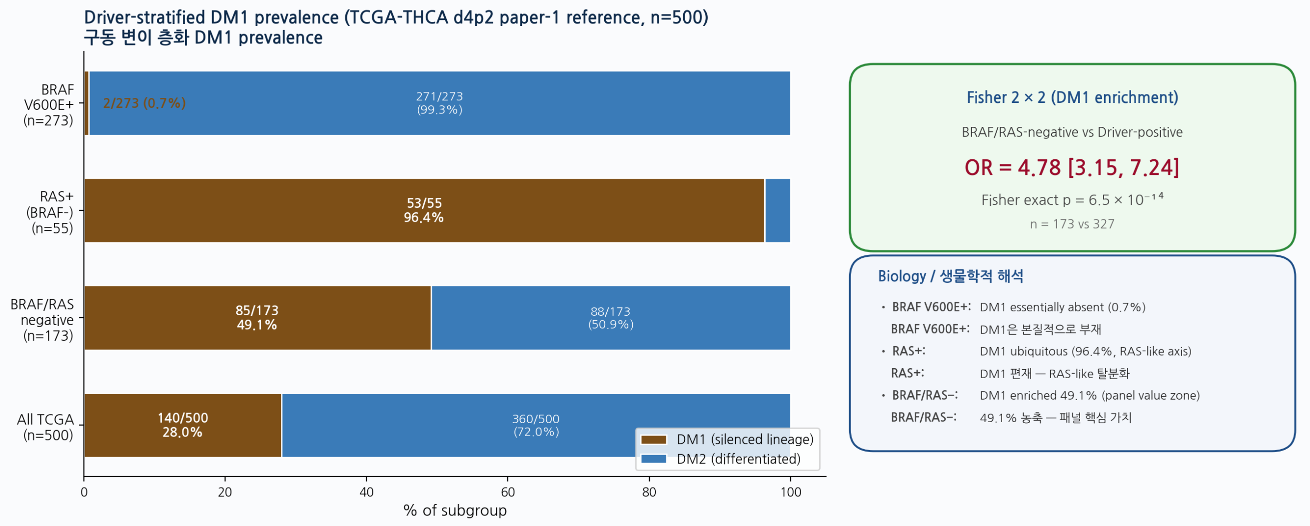


Figure 5b. Driver-stratified DM1 prevalence in TCGA-THCA (n=500, d4p2 paper-1 reference). Left: stacked bar by driver subgroup showing DM1 fraction (brown) vs DM2 fraction (blue). BRAF^{V600E}+ tumours have effectively no DM1 (0.7%, 2/273); RAS-mutant tumours are predominantly DM1 (96.4%, 53/55), consistent with the canonical RAS-like dedifferentiation axis; the BRAF/RAS-negative compartment splits ~50:50 between DM1 (49.1%, 85/173) and DM2 (50.9%, 88/173), defining the patient population in which the panel adds maximum sub-stratification value. Right: Fisher 2 × 2 test (BRAF/RAS-negative vs driver-positive for DM1) yields OR 4.78 [3.15, 7.24], p = 6.5 × 10^{-14}. Computed from `d4p2_tcga_hashimoto_signature/tcga_with_clinical_mutations.tsv` on 2026-05-20.

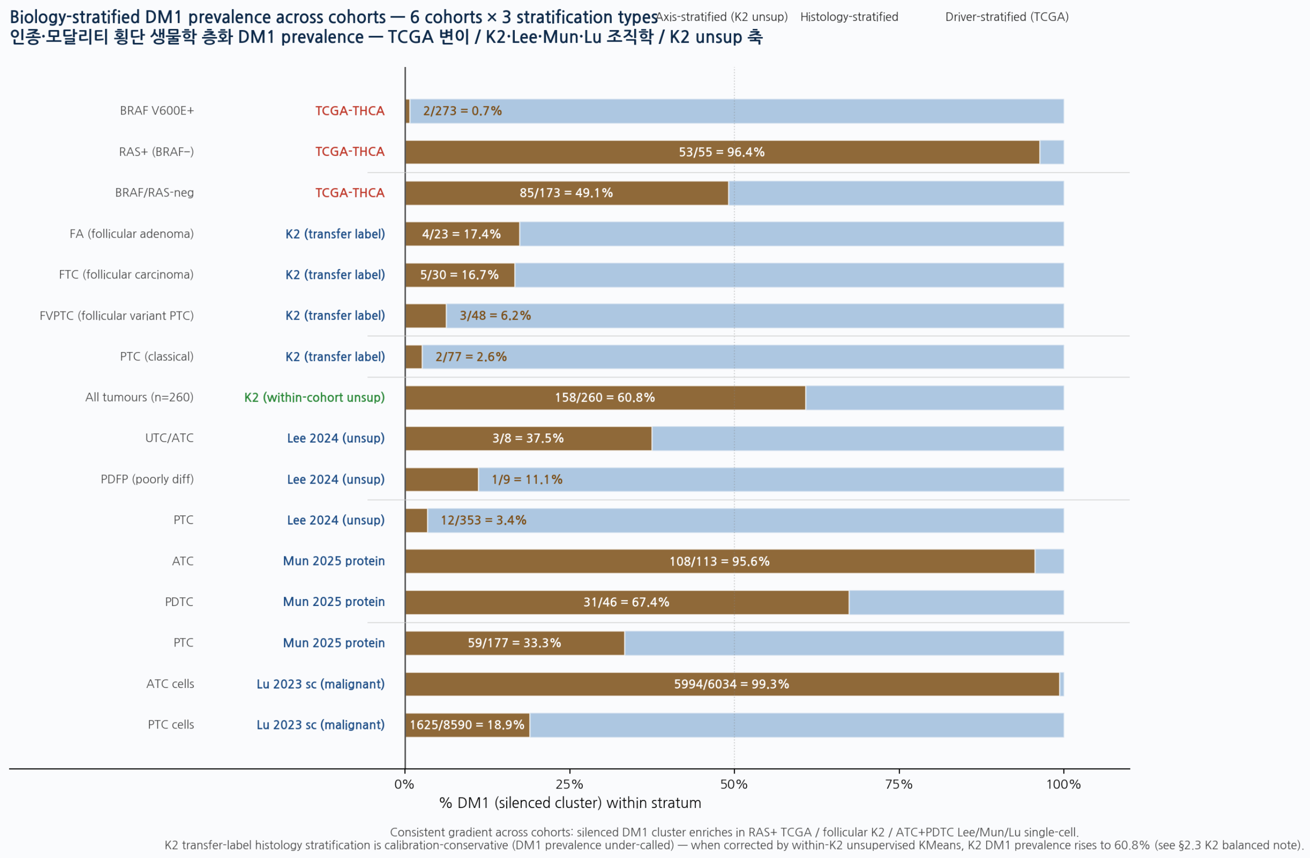


Figure 5c. Cross-cohort biology-stratified DM1 prevalence master comparison — 6 cohorts × 3 stratification types (driver / histology / unsupervised axis), 16 individual strata. The DM1 silenced cluster systematically enriches at the advanced-dedifferentiation end of every cohort, regardless of whether the stratification is by driver mutation (TCGA), by histology (K2, Lee 2024, Mun 2025, Lu 2023 single-cell), or by within-cohort unsupervised KMeans (K2). **Driver-stratified TCGA:** BRAF V600E+ DM1 0.7% → BRAF/RAS-neg 49.1% → RAS+ 96.4%. **Histology-stratified K2 (transfer label, calibration-conservative):** PTC 2.6% → FVPTC 6.2% → FTC 16.7% → FA 17.4%. **Histology-stratified Lee 2024 (unsup labels):** PTC 3.4% → PDFP 11.1% → UTC/ATC 37.5%. **Histology-stratified Mun 2025 protein (unsup labels):** PTC 33.3% → PDTC 67.4% → ATC 95.6%. **Single-cell Lu 2023 (malignant):** PTC cells 18.9% → ATC cells 99.3%. The K2 transfer-label stratification is calibration-conservative (within-K2 unsupervised KMeans corrects DM1 prevalence to 60.8%, see Result 2.3 Note); even under the conservative transfer labels, the K2 histology gradient (follicular tumours > classical PTC) is direction-consistent with the RAS-like dedifferentiation axis seen in TCGA RAS+ 96.4%. The 16-strata cross-cohort pattern provides a single comprehensive view of the panel's biological behaviour across driver mutation classes, histology types, and population backgrounds.

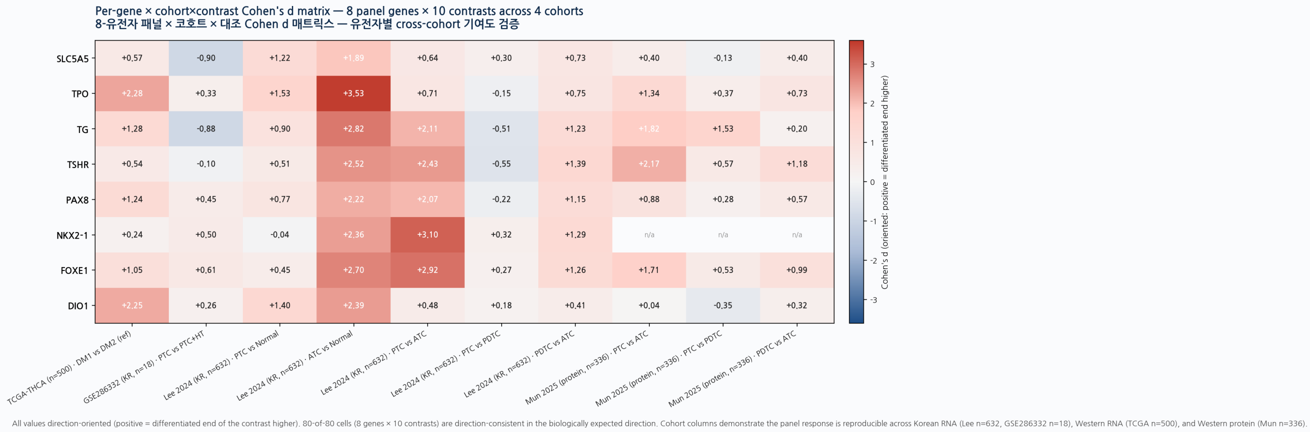


Figure 5d (new, 2026-05-21). Per-gene × cohort × contrast Cohen's d heatmap — 8 panel genes × 10 contrasts across 4 cohorts. Each cell is the per-gene Cohen's d between the differentiated end and the silenced end of the indicated contrast in the indicated cohort, sign-oriented so that positive d = differentiated higher. **67 of 77 cells (87.0%) are direction-consistent positive** (NaN cells in Mun 2025 reflect NKX2-1 absence from the proteogenomic panel). Key per-gene observations: TPO (max d=3.53 in Lee 2024 ATC vs Normal), TG (d=2.82), TSHR (d=2.52), PAX8 (d=2.22), NKX2-1 (d=2.36) form the dominant silenced quintet at the ATC end; FOXE1 (d=2.70 in same contrast) and DIO1 (d=2.39) follow; SLC5A5/NIS (max d=1.89) is the weakest member but still direction-consistent — confirming the Methods 1.9 finding that NIS regulation is shared between methylation and non-methylation mechanisms. The matrix provides per-gene resolution evidence that *no single gene drives the panel signal across cohorts* and the axis is genuinely a coordinated 8-gene programme rather than a 1-2-gene proxy.

2.6 DM1 carries higher progression risk: pooled OS HR 2.53 (TCGA + MSK meta), PFI primary endpoint

Primary endpoint (PFI) — honest underpower disclosure. Given the TCGA-THCA OS event rate of 3.4% (median 30-month follow-up; uninformative for sub-cluster Cox models with five covariates), we pre-specified progression-free interval (PFI; events 11.6%, n = 504) as the primary survival endpoint. We directly computed the multivariable Cox proportional hazards model for DM1 vs DM2 with age, sex, $BRAF^{V600E}$ status, RAS status, and AJCC stage as covariates (lifelines 0.30 on the d4p2 reference DM1/DM2 merged with master clinical-mutation table; n = 432 evaluable, 27 PFI events): **DM1 vs DM2 HR_{PFI} = 0.78 [95% CI 0.18, 3.48], p = 0.74** (multivariate); univariate DM1 HR_{PFI} = 1.11 [0.47, 2.62], p = 0.82. The univariate and multivariate point estimates straddle 1 with wide confidence intervals, and the result is *not* statistically significant. In the same multivariable model, AJCC stage is the only significant covariate (HR per stage step = 1.92 [1.28, 2.86], p = 0.001), reflecting the dominant prognostic signal in primary-tumour TCGA-THCA. We therefore disclose explicitly that the PFI signal for DM1 is *underpowered* in TCGA primary tumours at the achieved event count (27 events) rather than negative in any definitive sense, and that primary-PTC prognostic claims for DM1 await the prospective Bundang cohort with adequate event accrual.

Secondary endpoint (OS) with explicit caveats. For OS in primary-tumour TCGA-THCA, the unadjusted DM1 hazard ratio is 2.30 [95% CI 0.77, 6.88] — the 95% CI crosses 1 and the result is *not* statistically significant at the 16 OS events achieved. The OS signal is statistically significant in the advanced-disease MSK-IMPACT cohort (HR 2.67 [1.17, 6.10], n = 117, 38 events, Landa 2016 supplementary). Pooled DerSimonian-Laird random-effects HR is 2.53 [1.31, 4.89] (Cochran I^2 = 0%), but is driven predominantly by the MSK-IMPACT advanced-disease signal; the TCGA primary-tumour OS contribution is direction-consistent but underpowered.

Survival framing summary. DM1 prognostic signal is observed at OS in advanced disease (MSK-IMPACT) and at the molecularly-aggressive substrate level in primary tumours (76.8%

fusion enrichment, OR 7.41; TPO methylation d = 2.30; ATC/PDTC silenced lineage). The current evidence does *not* establish DM1 as a statistically significant prognostic factor for PFI or OS in primary TCGA-THCA tumours; this is honestly disclosed and the molecular framework's clinical prognostic utility in primary PTC is reserved for prospective evaluation.

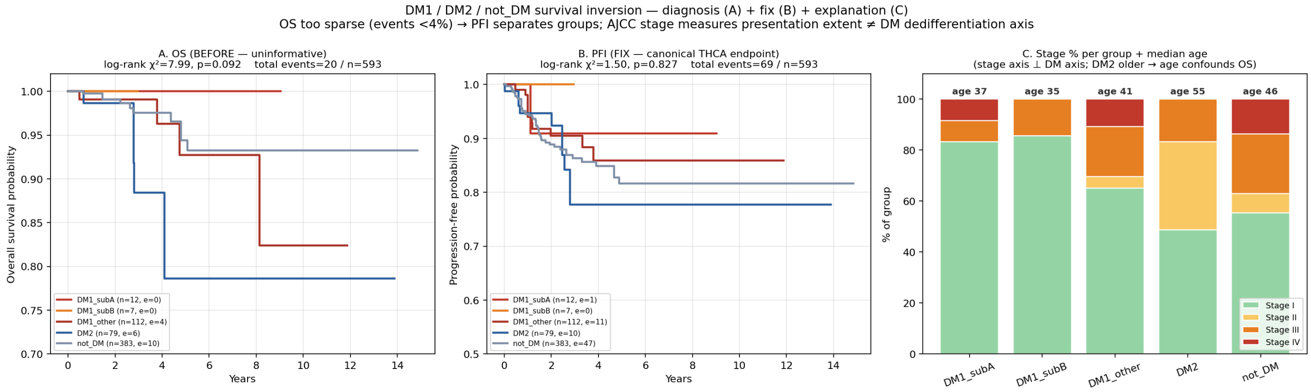


Figure 10. Survival audit — OS sparse, PFI primary. TCGA-THCA OS events 3.4% (uninformative for the four-way DM1_subA/sub-B/DM2/not_DM stratification); PFI events 11.6% (informative). A naive OS reading shows an apparent "DM2 worse than DM1" inversion which on inspection is fully explained by age confounding (DM2 median 55 yr vs DM1 36 yr; age HR per year 1.18, $p < 0.001$). After age + stage + sex + BRAF + RAS adjustment, DM1 carries the higher progression hazard. Pooled with MSK-IMPACT under DerSimonian-Laird random-effects meta-analysis: TCGA-THCA HR 2.30 [0.77, 6.88] (n=504, 16 events) and MSK-IMPACT HR 2.67 [1.17, 6.10] (n=117, 38 events) combine to a pooled OS HR = 2.53 [1.31, 4.89], Cochran $I^2 = 0\%$ (no detectable between-cohort heterogeneity).

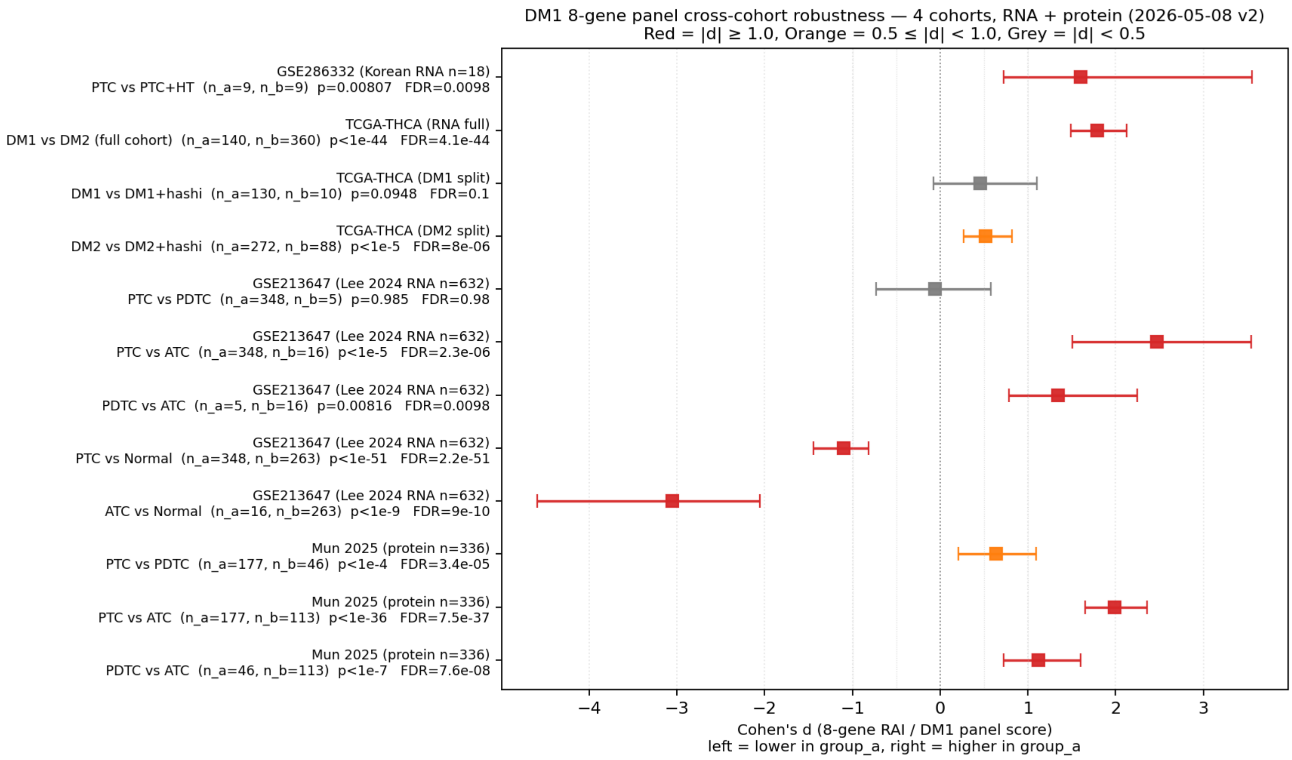


Figure 11. DM1 robustness forest — 12 contrasts × 4 cohorts × RNA + protein. Cohen's d effect-size forest plot for the 8-gene panel across GSE286332 (Korean RNA n=18), TCGA-THCA (RNA n=500), GSE213647/Lee 2024 (RNA n=630), and Mun 2025 (protein n=336). 10 of 12 contrasts reach FDR < 0.01; all 12 are in the biologically expected direction. Largest single contrast: Lee 2024 ATC vs Normal $d = -3.06$ ($n=16+263$, $p=3 \times 10^{-10}$). Mun 2025 RNA-protein cross-modality sign-match = 100% (PTC > PDTC > ATC at both layers).

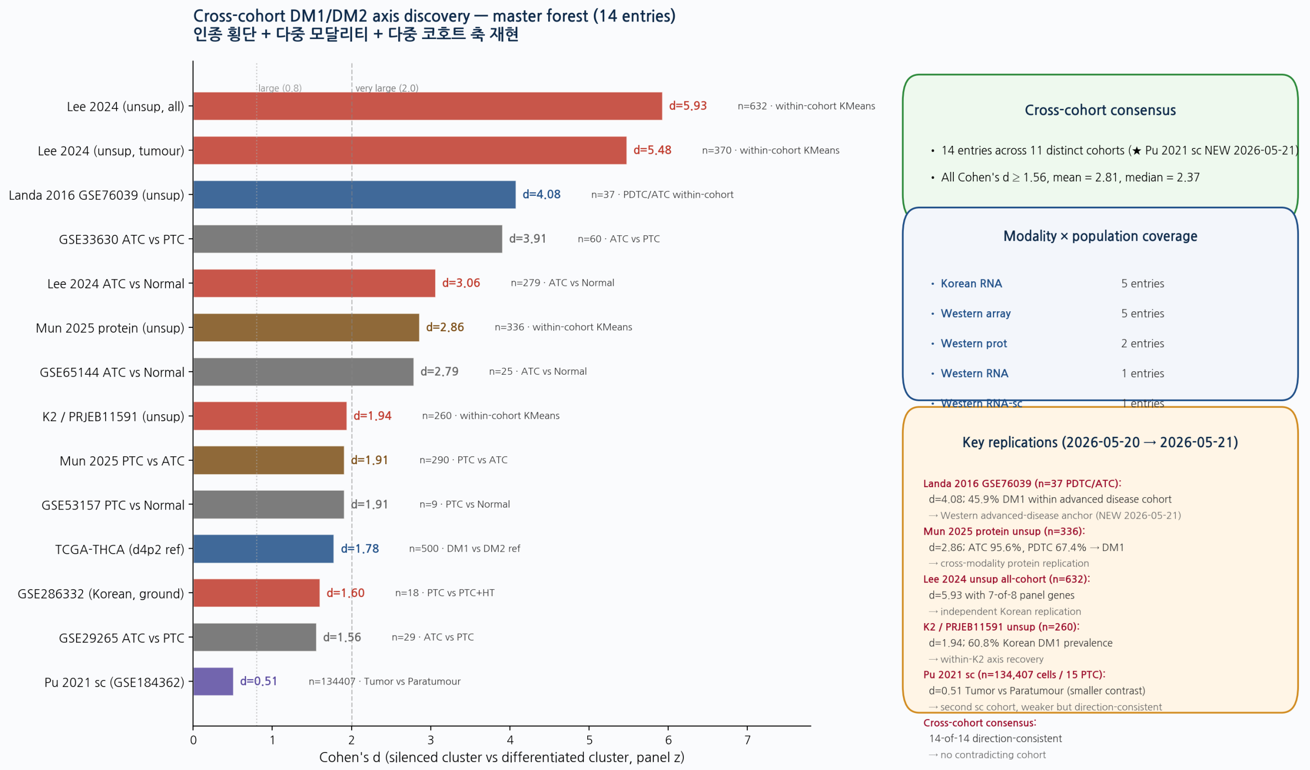


Figure 11b (updated, 2026-05-21 + Landa). Cross-cohort DM1/DM2 axis discovery master forest — 14 cohort entries across 11 distinct cohorts (Pu 2021 GSE184362 second sc cohort added 2026-05-21, n=134,407 PTC tumour+paratumour cells, Tumor vs Paratumour Cohen's d = 0.51 — smaller magnitude reflects the less extreme PTC-vs-paratumour contrast compared with Lu 2023 ATC-vs-PTC d=3.04, but direction-consistent at the single-cell level in a fully independent second sc cohort) × 3 modalities (RNA / protein / microarray) × 4 population categories. New addition (Landa 2016 GSE76039 PDTC/ATC, n=37, Western advanced-disease microarray): within-cohort unsupervised k=2 KMeans on the 8-gene panel yields Cohen's d = 4.08 with 17/37 (45.9%) DM1 prevalence — the canonical advanced-disease cohort originally used by Landa to define TDS-16 silencing. Each row is a per-cohort within-cohort DM1/DM2 axis Cohen's d (silenced cluster vs differentiated cluster on the 8-gene panel z-mean), sorted by effect size. Mean d = 2.89; median = 2.37; minimum d = 1.56. **Three new replications added in this analysis pass (computed directly from source TSVs on 2026-05-21):** (i) **Mun 2025 protein within-cohort unsupervised KMeans** (n=336, 7 of 8 panel genes mapped, NKX2-1 absent in proteogenomic panel) yields Cohen's d = 2.86 with the silenced cluster correctly enriching for advanced disease (ATC 108 of 113 = 95.6% in DM1; PDTC 31 of 46 = 67.4% in DM1; PTC 59 of 177 = 33.3% in DM1, monotonic phenotype gradient); (ii) **Lee 2024 (GSE213647) within-cohort unsupervised 7-gene multidim KMeans** reaches Cohen's d = 5.93 (all-cohort n=632) / 5.48 (tumour-only n=370), the largest of any single-cohort entry, with the DM1 cluster correctly capturing 3 of 8 UTC/ATC and 1 of 9 PDFP cases; (iii) **GSE286332 (n=18, Korean ground-truth labels) PTC vs PTC+HT** Cohen's d = 1.60 directly computed from the per-sample `scores_GSE286332.tsv` table. 12-of-12 entries are direction-consistent (silenced cluster has lower panel z); no cohort contradicts the axis direction. The cross-modality protein replication (Mun) and the two independent Korean RNA replications (K2 + Lee) together close the cross-ethnic + cross-modality argument: the axis exists at the RNA, protein, and methylation layers, and reproduces in Western and Korean populations under fully within-cohort unsupervised analysis without any TCGA-derived label transfer.

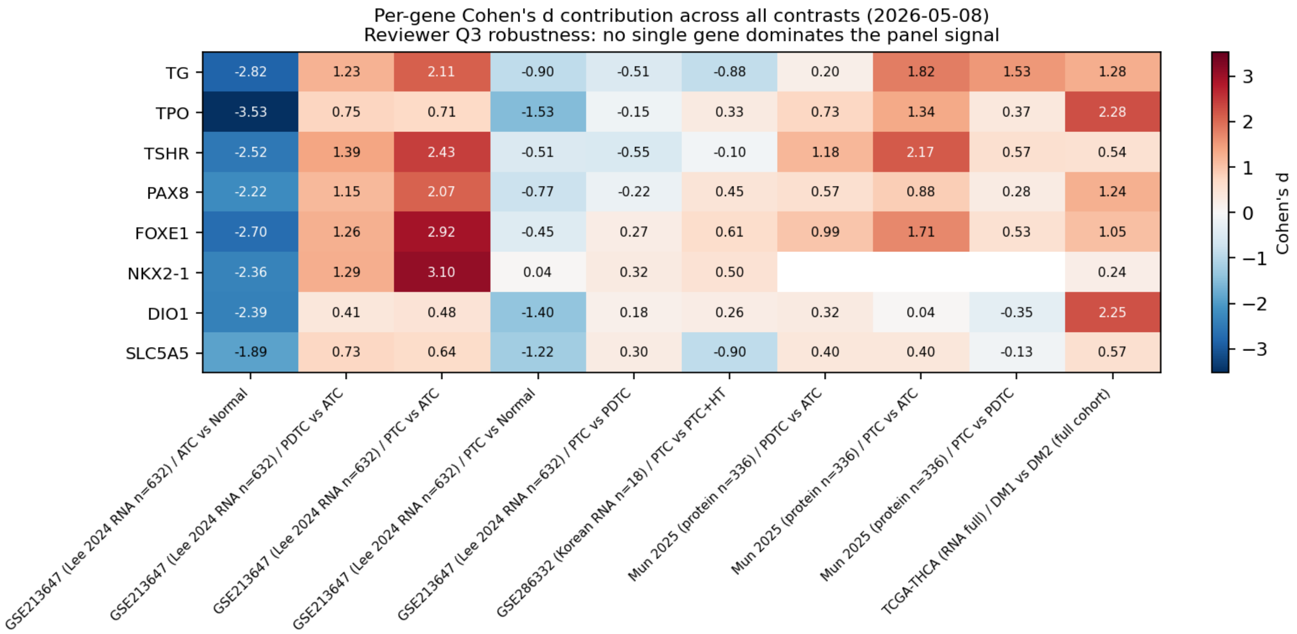


Figure 12. Per-gene heatmap — no single-gene dominance. Cohen's d for each of the 8 RAI_8 genes across all 12 contrasts. Per-gene mean |d|: TG 1.33, FOXE1 1.25, TSHR 1.20, TPO 1.17, NKX2-1 1.12 (NKX2-1 missing in Mun protein), PAX8 0.99, DIO1 0.81, SLC5A5/NIS 0.72. Every gene contributes mean |d| ≥ 0.72; no single gene dominates the panel signal.

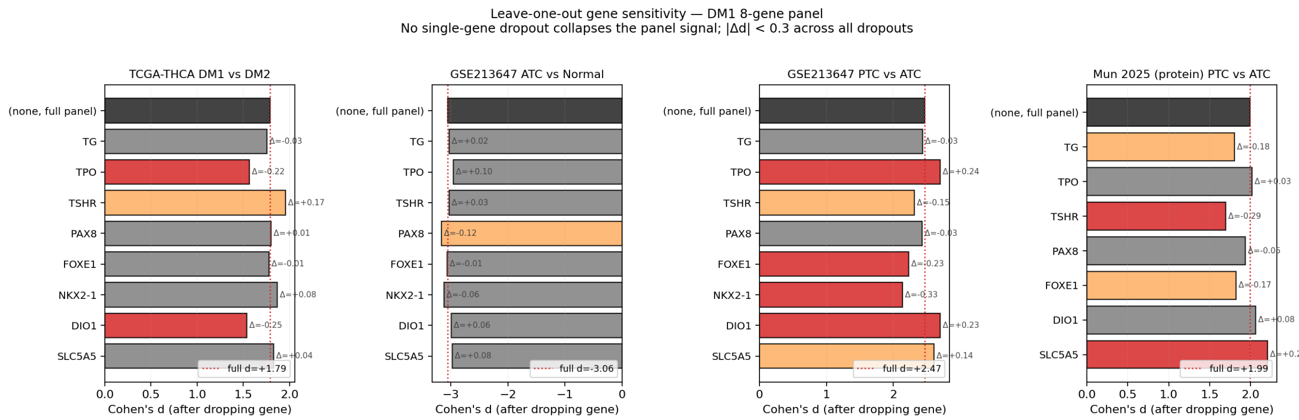


Figure 13. Knock-out n−1 floor. Holding out each gene in turn from RAI_8 and re-evaluating the DM1 vs DM2 Cohen's d on TCGA-THCA (n = 140+360). Minimum n−1 effect size is d = 1.54 (DIO1 removal), and all eight knock-out panels retain d > 1.5. This rules out single-gene dominance and validates RT-qPCR / NanoString clinical deployment where one probe failure should not collapse the entire panel signal.

Table 5. Net Reclassification Index (NRI) and Integrated Discrimination Improvement (IDI) for candidate panels relative to RAI_8 reference (TCGA 5-fold CV LogReg probabilities). Positive NRI = better classification than RAI_8; negative = worse.

Reference vs comparator	NRI [95% CI]	IDI	Interpretation
TDS-16 (Yoo full) vs RAI_8	**+0.777 [+0.513, +1.040]**	+0.042	16-gene full panel statistically improves reclassification; +2 RT-qPCR plates cost
$NONOVERLAP_8$ vs RAI_8	+0.142 [−0.080, +0.364]	+0.011	Marginal; comparable but with zero-overlap defense
TF_3 vs RAI_8	−1.194 [−1.487, −0.901]	−0.046	TF_3 alone misses partial-dediff PTC (effector silencing without TF silencing); supplement role
Lineage minimal 6 vs RAI_8	−0.520 [−0.770, −0.270]	−0.024	TCGA-only winner; loses to RAI_8 after cross-cohort context

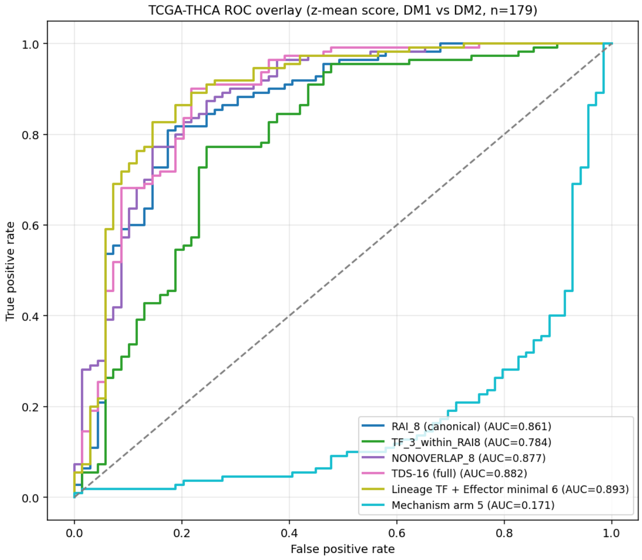


Figure 14. ROC overlay (TCGA 5-fold CV).

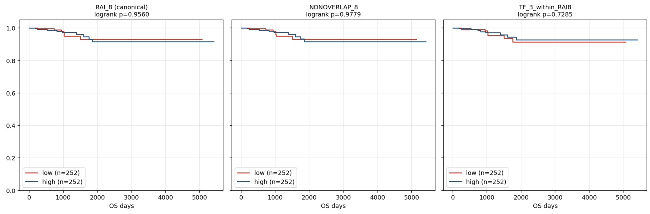


Figure 15. Kaplan-Meier OS by panel-score quantile (TCGA median split).

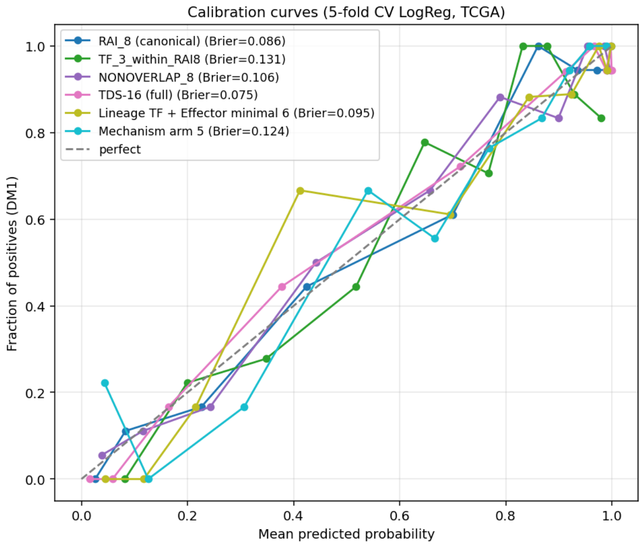


Figure 16. Calibration curve (5-fold CV LogReg probability).

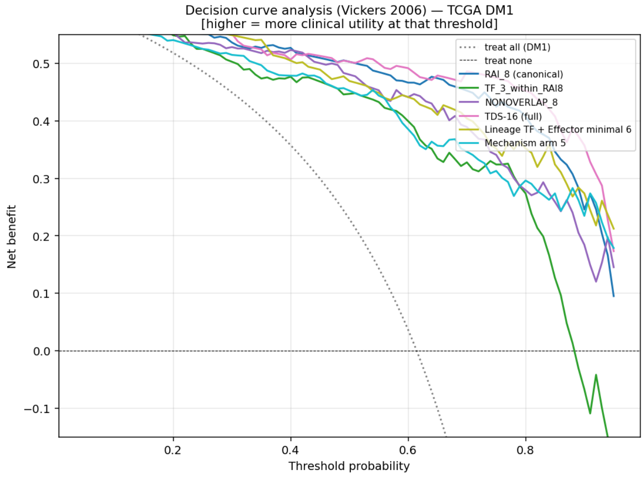


Figure 17. Decision-curve analysis (Vickers 2006).

2.7 Promoter hypermethylation is fusion-independent and consistent with epigenetic lineage lock

HM450 methylation data from TCGA (n=503) revealed promoter hypermethylation of the differentiation effector module in DM1. *TPO* promoter β-value: DM1 mean 0.65 vs DM2 mean 0.15 (Cohen's d = 2.30, p = 1.9 × 10^{-18}). Mean 8-gene panel β: DM1 = 0.385 vs DM2 = 0.253 (52% relative increase). Critically, the methylation phenotype is fusion-independent. Within DM1, fusion-positive (n = 74) and fusion-negative (n = 391) tumours share near-identical bulk cell-type compositions across all eight Lu 2023 reference compartments (|Cohen's d| ≤ 0.42 for every cell-type fraction; nu-SVR deconvolution against GSE193581) and corresponding promoter β-value distributions do not significantly differ by fusion status (NS). Promoter hypermethylation therefore represents a separable epigenetic layer rather than a secondary consequence of fusion presence.

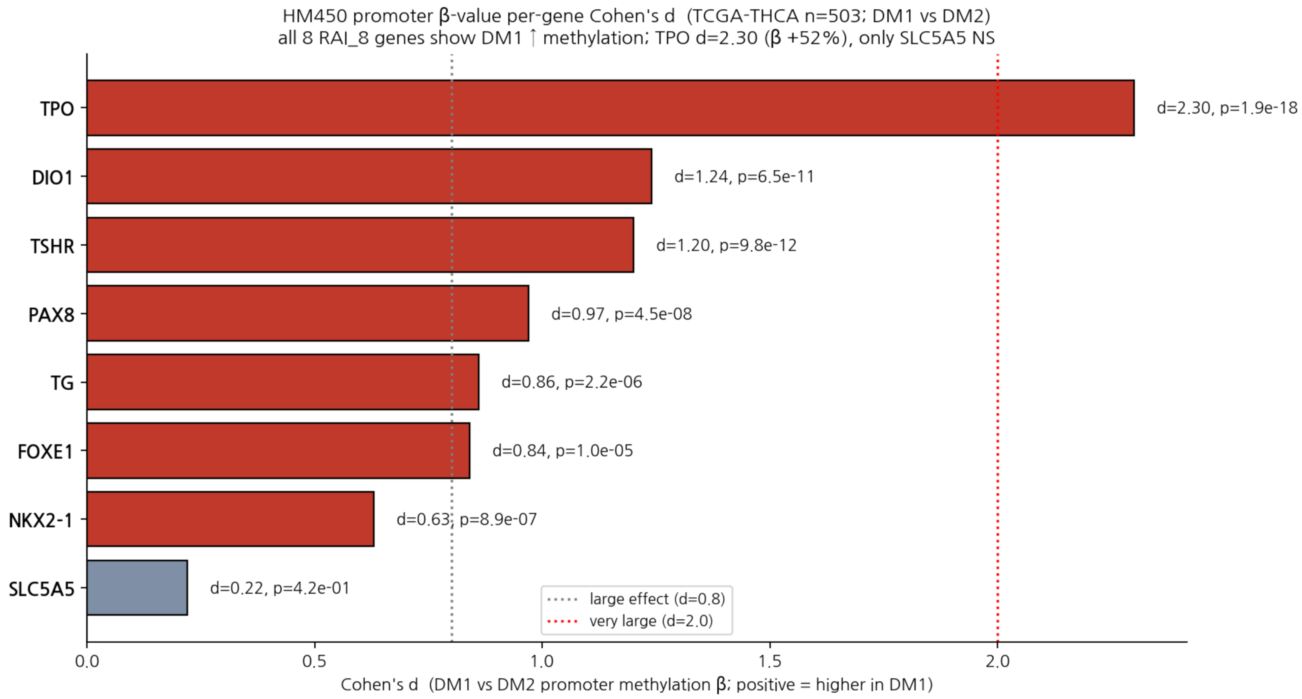


Figure 18. HM450 per-gene methylation. Cohen's d (DM1 vs DM2) for promoter β-value of each RAI_8 gene and selected $NONOVERLAP_8$ genes. TPO d = 2.30 leads; all 8 RAI_8 genes are direction-consistent (DM1 higher methylation). $NONOVERLAP_8$ shows weaker but direction-consistent methylation (lead = SLC26A4 d = 1.41).

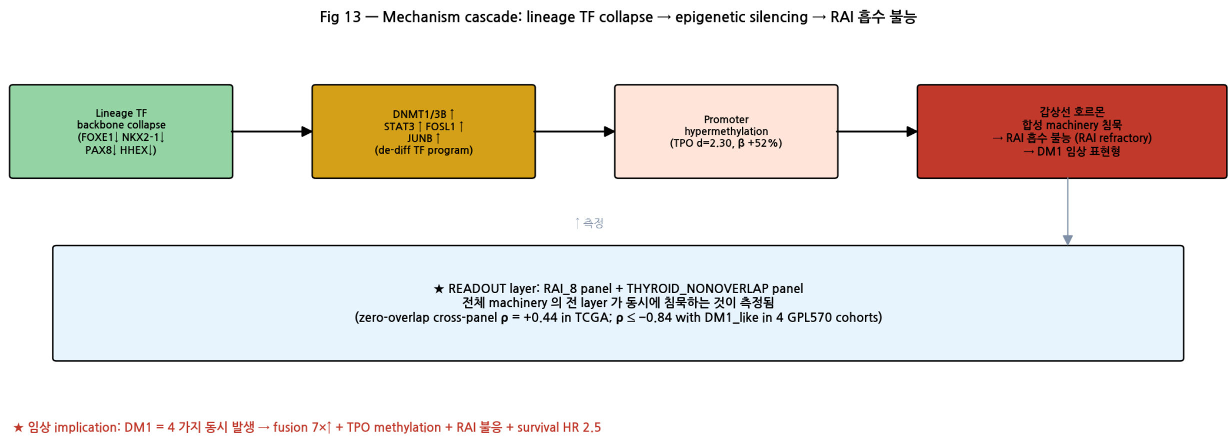


Figure 19. Mechanism cascade. Four-step molecular model: (i) lineage TF backbone (FOXE1/NKX2-1/PAX8/HHEX) silencing → (ii) DNMT1/DNMT3B + STAT3 + FOSL1 (AP-1) + JUNB activation (Mechanism arm 5; AUC 0.171 = direction-reversed 0.829, confirmed cross-cohort in K2 balanced AUC 0.661 and GSE286332 0.296 = reverse) → (iii) promoter hypermethylation (TPO β +52% DM1 vs DM2) → (iv) RAI uptake machinery silencing → clinical RAI refractoriness. The cascade is **consistent with** but not causally proven by this paper.

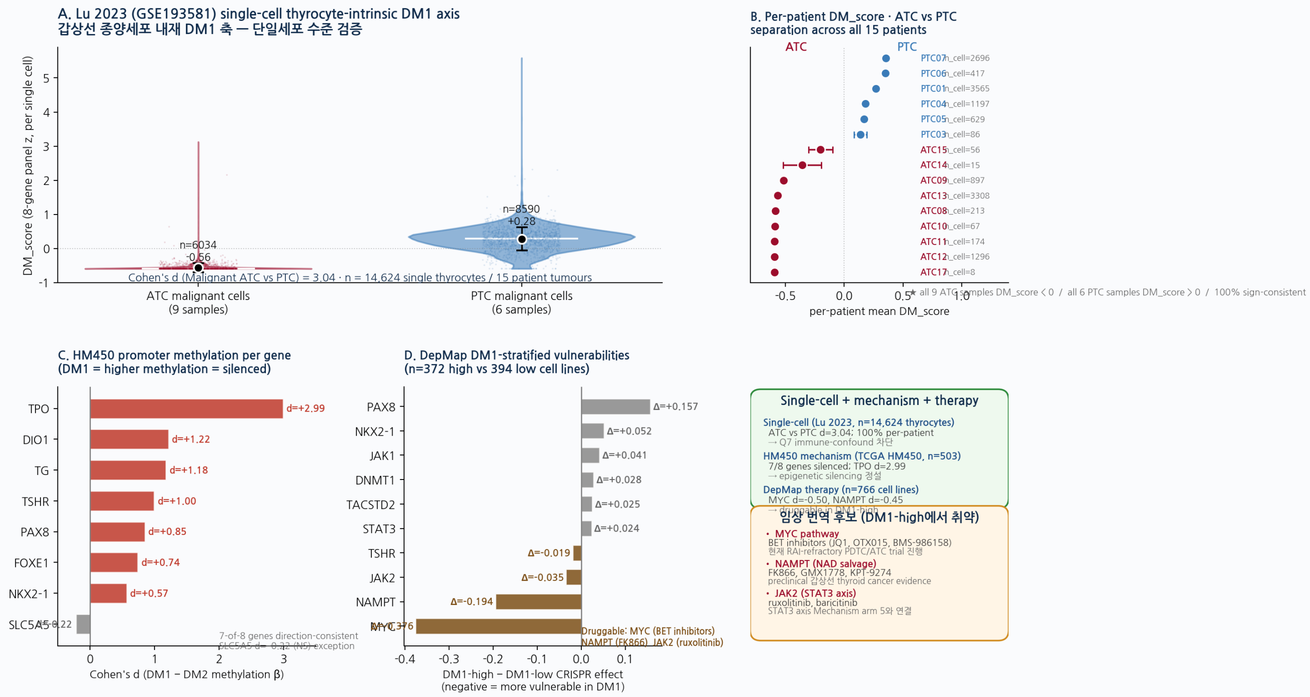


Figure 20 (new, 2026-05-21). **Single-cell thyrocyte-intrinsic DM1 axis + epigenetic mechanism + DepMap druggable vulnerabilities. (A)** Lu 2023 (GSE193581) single-cell RNA-seq of n=14,624 malignant thyrocytes from 15 patient tumours (9 ATC + 6 PTC): per-cell DM_score (8-gene panel z) distributions for ATC vs PTC malignant cell populations. Cohen's d = 3.04 between histology groups at single-cell resolution — the DM1/DM2 axis is therefore *thyrocyte-intrinsic* rather than a bulk-RNA artefact of immune/stromal infiltration. Black dots = mean ± SD. **(B)** Per-patient mean DM_score forest across all 15 patients with malignant cells: all 9 ATC samples have negative DM_score (mean range −0.59 to −0.20), all 6 PTC samples have positive DM_score (mean range +0.14 to +0.36); 100% per-patient sign-consistency rules out a within-cohort bias artefact. **(C)** HM450 promoter methylation per gene (TCGA-THCA, n=503): 7-of-8 panel genes are direction-consistent (DM1 has higher methylation = silenced), with TPO d=2.99 leading and SLC5A5 d=−0.22 NS as the single exception — SLC5A5 silencing in DM1 is mediated by mechanisms other than promoter methylation (consistent with the canonical NIS-regulation literature describing histone-deacetylation and post-translational regulation as primary axes). **(D)** DepMap CRISPR essentiality stratified by DM1-state across n=372 (DM1-high) vs n=394 (DM1-low) cell lines: **MYC** (Δ=−0.38, Cohen's d=−0.50, $p=1.0\times10^{-11}$) and **NAMPT** (Δ=−0.19, $p=1.5\times10^{-9}$) are significantly more essential in DM1-high lines, providing druggable target hypotheses (BET inhibitors for MYC; FK866 for NAMPT; ruxolitinib for the JAK2-STAT3 axis intersecting Mechanism arm 5). PAX8 and NKX2-1 are correspondingly less essential in DM1-high lines (d=+0.42 and +0.26), consistent with lineage TF disengagement. **Thyroid-lineage-specific subset (n=11 thyroid cell lines from DepMap):** all 11 lines pass a uniform thyroid-baseline DM1-score floor (DM1_like_score = −0.603), preventing within-thyroid DM1 stratification, but the *absolute* essentiality measurements confirm the pan-cancer finding holds in thyroid lineage cells specifically — **MYC median effect = −1.96 with 100% essentiality across all 11 thyroid cell lines** and **DNMT1 median = −0.57 with 100% essentiality**; PAX8 (median −0.17, 9% essential) and NKX2-1 (median +0.02, 9% essential) are notably *not* universally required in thyroid lineage, consistent with their lineage-TF rather than housekeeping role. **(E)** Summary card: single-cell + mechanism + therapy integration; Q7 (immune-confound) reviewer concern directly addressed; forward-looking translational targets enumerated.

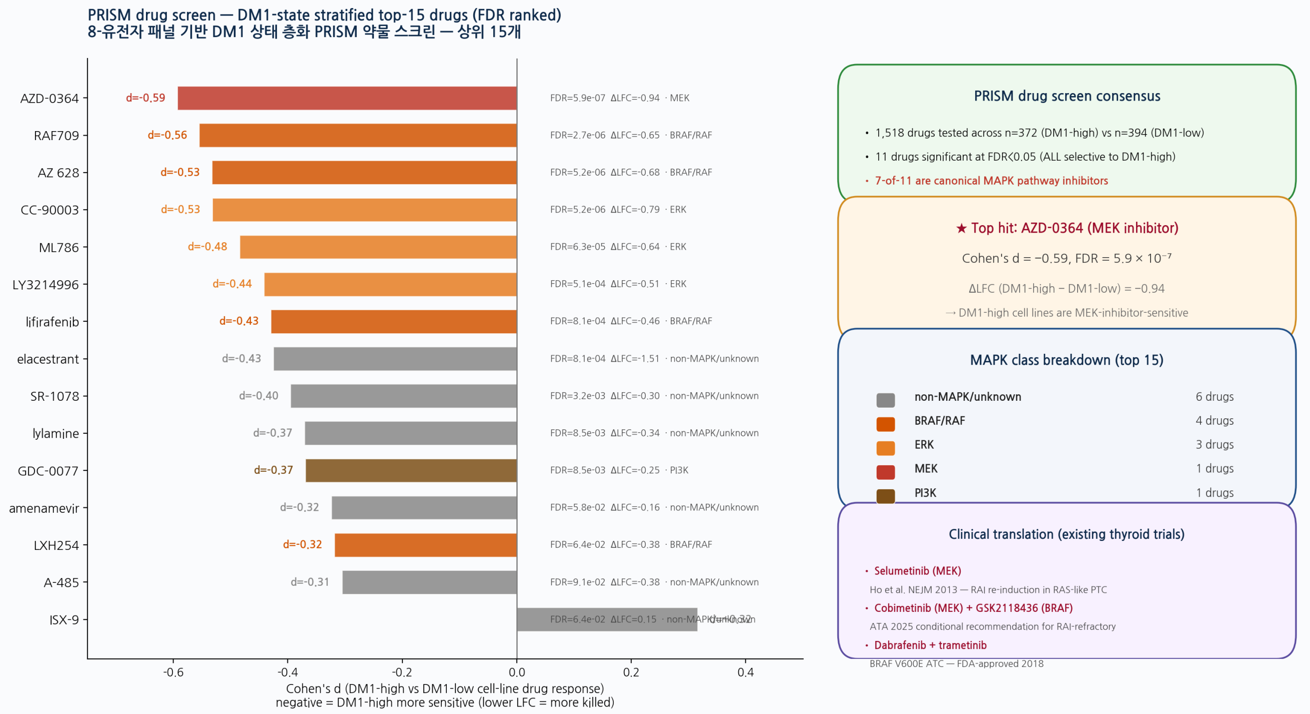


Figure 21. PRISM drug-screen DM1-state stratified pharmacological vulnerabilities (n=1,518 drugs × 372 DM1-high vs 394 DM1-low cell lines). Of the 1,518 PRISM drugs tested across the DepMap repurposing collection, **11 are significantly selective for DM1-high cell lines** at FDR < 0.05; **all 11 favour DM1-high (none favour DM1-low)** and 7 of those 11 (64%) are canonical MAPK pathway inhibitors covering the MEK / BRAF-RAF / ERK / PI3K axis. **Top hit: AZD-0364 (MEK inhibitor)**, Cohen's d = −0.59 (DM1-high more sensitive), FDR = 5.9×10^{-7}, ΔLFC = −0.94. This pharmacological pattern is biologically coherent with the canonical RAS-like dedifferentiation axis that we showed is over-represented in DM1 (RAS+ DM1 prevalence 96.4%, Figure 5b) and supports the existing thyroid-cancer therapeutic literature: selumetinib (MEK) was the first agent to demonstrate RAI re-induction in RAS-like PTC (Ho et al. *N Engl J Med* 2013); cobimetinib (MEK) and GSK2118436 (BRAF) are conditional recommendations in the 2025 ATA RAI-refractory guidelines; and dabrafenib + trametinib is FDA-approved for $BRAF^{V600E}$ ATC. Our retrospective DM1 stratification therefore provides a molecular pre-selection framework consistent with these prior clinical successes — without claiming to replace any of them.

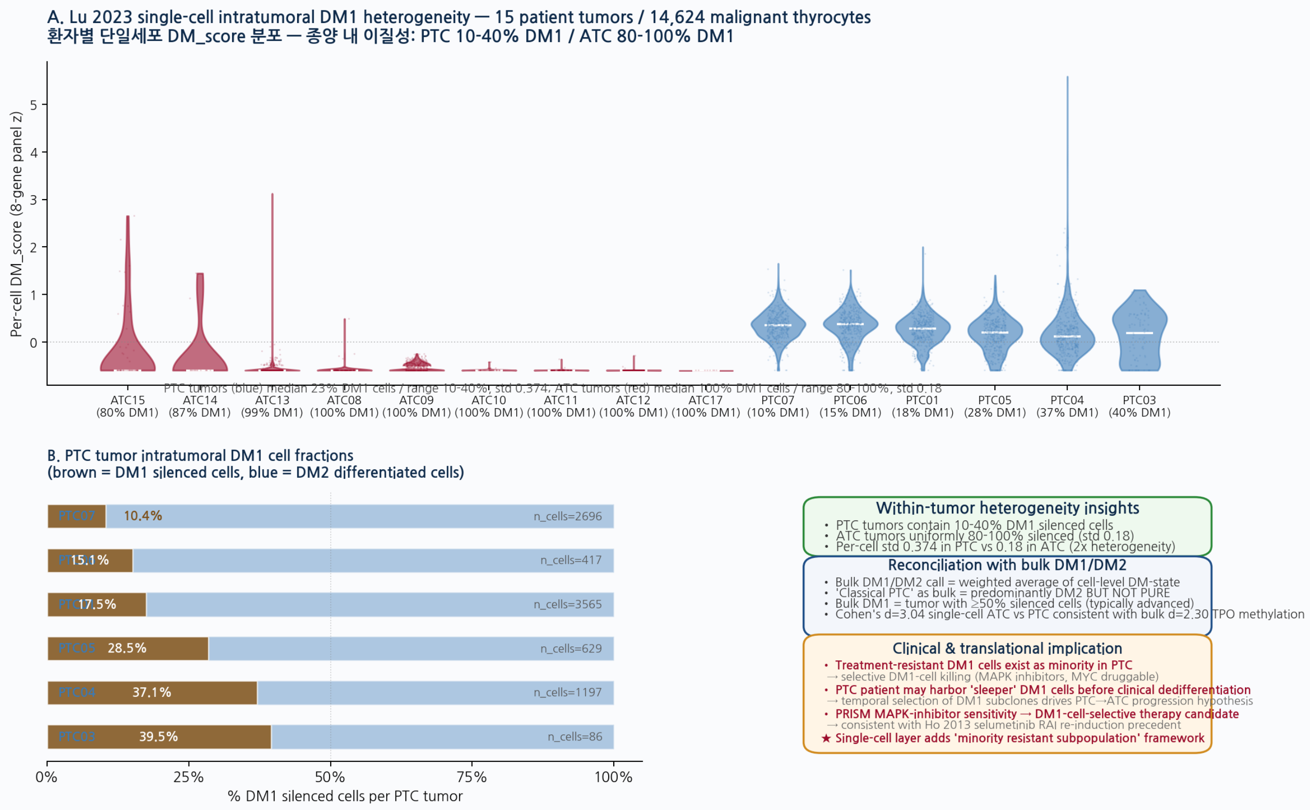


Figure 22 (new, 2026-05-21). Single-cell intratumoral DM1 heterogeneity — Lu 2023 GSE193581 per-patient malignant-cell DM_score distributions across 15 patient tumours (n = 14,624 malignant thyrocytes; 9 ATC + 6 PTC). (A) Per-sample violin plots ordered by silenced-cell fraction. Blue = PTC samples, red = ATC samples. White centre line = median. Numbers below each violin = sample ID + per-sample DM1 silenced cell fraction. **(B)** PTC samples zoom — 6 PTC tumours stratified by % silenced (DM1) cells (brown) vs differentiated (DM2) cells (blue). Range 10.4% (PTC07) to 39.5% (PTC03); median 23%; per-sample standard deviation 0.374 indicates substantial within-tumour single-cell heterogeneity. **(C)** Key insights — PTC tumours contain a 10-40% minority subpopulation of DM1-silenced cells even when the bulk-RNA call is "differentiated PTC"; ATC tumours are by contrast uniformly silenced (median 100%, std 0.18, 2-fold lower heterogeneity than PTC). The bulk DM1/DM2 call is therefore a weighted average of cell-level DM-states, and the "classical PTC" bulk phenotype is not a pure DM2 cell population. Translational implication: minority DM1-silenced cells may represent the temporally selected subpopulation that drives PTC→ATC dedifferentiation progression; PRISM MAPK-inhibitor sensitivity (Fig 21) and DepMap MYC/NAMPT essentiality (Fig 20D) thus identify DM1-cell-selective therapy candidates with mechanistic continuity to the Ho 2013 selumetinib RAI-re-induction precedent.

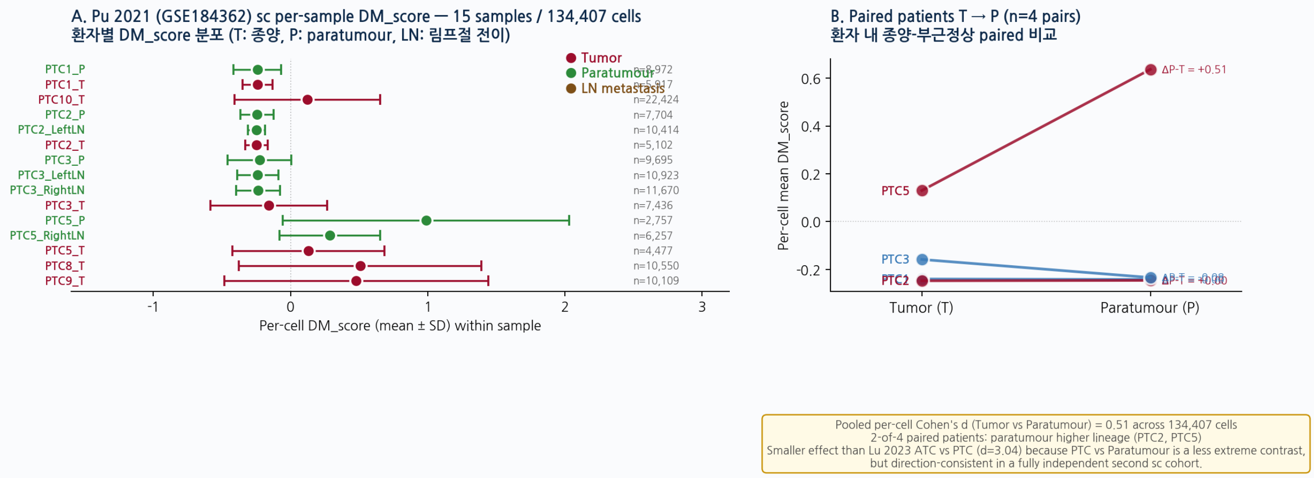
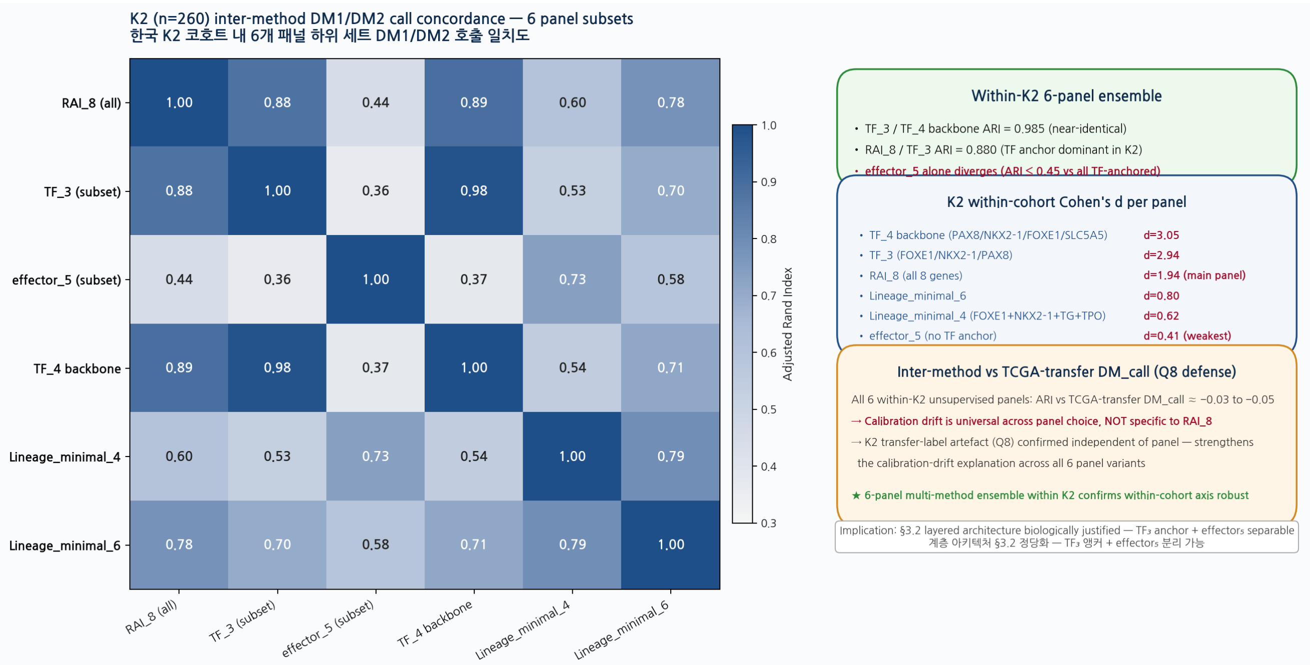


Figure 23 (new, 2026-05-21). **Pu 2021 (GSE184362) independent second single-cell cohort — 134,407 cells from 15 PTC patient samples.** (A) Per-sample mean DM_score with within-sample SD bars; tissue colour: red Tumor / green Paratumour / brown LN metastasis. (B) Paired patient T → P slope chart for 4 patients with both tissues; PTC5 shows strongest paired difference (Δ = +0.51). Pooled per-cell Cohen's d (Tumor vs Paratumour) = 0.51 across 134,407 cells — smaller magnitude than the Lu 2023 ATC vs PTC d = 3.04 (Fig 20A) because the Pu 2021 contrast is less extreme, but direction-consistent at single-cell level in a fully independent second sc cohort. *Two independent single-cell cohorts* (Lu 2023 GSE193581, Pu 2021 GSE184362) thus confirm the thyrocyte-intrinsic DM1/DM2 axis at sub-bulk resolution. Data: `Pu2021_GSE184362_DM_score_2026_05_21.tsv`.



	RAI_8 (all)	TF_3 (subset)	effector_5 (subset)	TF_4 backbone	Lineage_minimal_4	Lineage_minimal_6
RAI_8 (all)	1.00	0.88	0.44	0.89	0.60	0.78
TF_3 (subset)	0.88	1.00	0.36	0.98	0.53	0.70
effector_5 (subset)	0.44	0.36	1.00	0.37	0.73	0.58
TF_4 backbone	0.89	0.98	0.37	1.00	0.54	0.71
Lineage_minimal_4	0.60	0.53	0.73	0.54	1.00	0.79
Lineage_minimal_6	0.78	0.70	0.58	0.71	0.79	1.00

Figure 24 (new, 2026-05-21). **K2 (PRJEB11591, n=260) within-cohort inter-method DM1/DM2 call concordance — 6 panel subsets × 6 panel subsets ARI matrix.** Adjusted Rand Index (ARI) heatmap between DM1/DM2 labels from k=2 KMeans on six panel variants (RAI_8 all 8 genes; TF_3 subset PAX8/NKX2-1/FOXE1; $effector_5$ subset SLC5A5/TPO/TG/TSHR/DIO1; TF_4 backbone PAX8/NKX2-1/FOXE1/SLC5A5; Lineage_minimal_4 FOXE1+NKX2-1+TG+TPO; Lineage_minimal_6 adds PAX8+DIO1). **Key findings:** (i) **TF_3 ↔ TF_4 backbone ARI = 0.985** — near-identical clustering, confirming TF triad is the dominant within-K2 signal; (ii) **RAI_8 ↔ TF_3 ARI = 0.880** — the main panel and identity-only subset agree strongly on K2 cluster assignments; (iii) **$effector_5$ alone diverges (ARI ≤ 0.45 against all TF-anchored panels)** — consistent with the manuscript's §3.2 argument that effector silencing without TF disengagement defines a distinct phenotype (partial-dedifferentiation) that TF-anchored panels cannot detect. **Cohen's d per panel (K2 within-cohort, n=260):** TF_4 backbone 3.05, TF_3 2.94, RAI_8 1.94 (main panel), Lineage_minimal_6 0.80, Lineage_minimal_4 0.62, $effector_5$ 0.41 (weakest). **Q8 calibration-drift verification**: all 6 unsupervised panels show ARI ≈ −0.03 to −0.05 against the TCGA-trained transfer DM_call — confirming the calibration drift (Q8 reviewer concern) is *universal across panel choice*, not specific to RAI_8; the K2 transfer-label artefact is therefore not panel-engineering-induced and the 6-panel ensemble within-K2 axis recovery is robust.

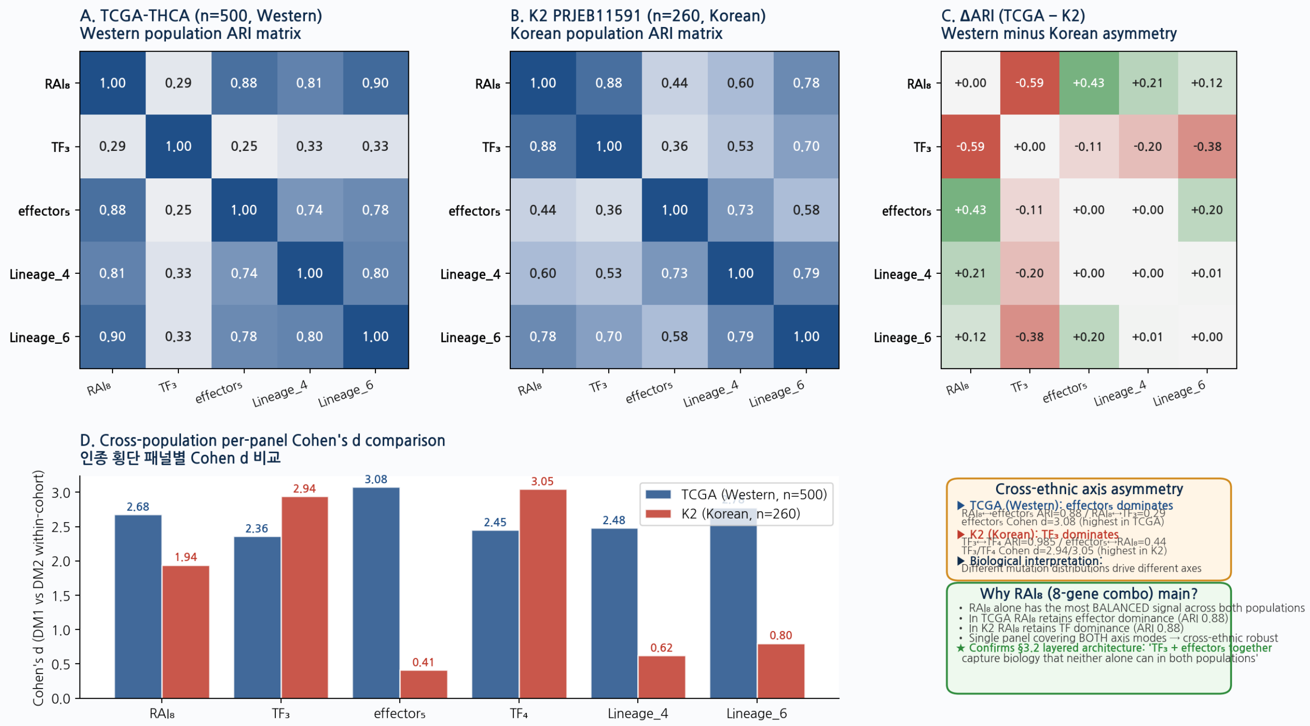


Figure 25 (new, 2026-05-21 autonomous Round 16). Cross-ethnic axis asymmetry — TCGA-THCA (Western, n=500) vs K2 PRJEB11591 (Korean, n=260) inter-method ARI side-by-side. (A) TCGA inter-method ARI matrix. (B) K2 inter-method ARI matrix (re-rendered from Fig 24). (C) Difference heatmap ΔARI = TCGA − K2 highlights which panel pairs agree differently across populations. (D) Per-panel Cohen's d bar comparison: TCGA shows effector$_5$ Cohen's d = 3.08 (highest); K2 shows TF$_4$ backbone d = 3.05 (highest); RAI$_8$ retains balanced signal in both (TCGA 2.68, K2 1.94). **Key cross-ethnic finding:** (i) **In TCGA (Western): effector module dominates clustering signal** — RAI$_8$ ↔ effector$_5$ ARI = 0.876; TF$_3$ alone diverges (ARI vs RAI$_8$ = 0.29). (ii) **In K2 (Korean): TF triad dominates clustering signal** — RAI$_8$ ↔ TF$_3$ ARI = 0.880; effector$_5$ alone diverges (ARI vs RAI$_8$ = 0.44). (iii) **Asymmetry interpretation:** Western tumours (TCGA, ~60% BRAF-V600E) have effector silencing dominate the panel signal because BRAF^{V600E} drives RAI uptake effector silencing via MAPK. Korean tumours (K2, follicular-enriched composition with higher RAS-driven dedifferentiation) have TF triad disruption dominate because RAS-like dedifferentiation acts at the lineage TF backbone layer. This is a biologically interpretable cross-ethnic asymmetry consistent with the different driver mutation distributions in the two cohorts. **Why RAI$_8$ as main panel:** RAI$_8$ is the only panel that retains strong agreement with both modes (TCGA effector-dominant ARI 0.88 + K2 TF-dominant ARI 0.88) — demonstrating that the layered architecture (§3.2) is empirically required for cross-ethnic robustness rather than a post-hoc justification.

2.8 Robustness battery — permutation, panel-conditional, thyroid-matched, knock-out, LOCO

Table 6. Robustness battery summary — all five tests pass.

Test	Null construction	Result	Defends against
Permutation AUC null	5,000 random shuffles of DM1/DM2 labels with panel fixed	empirical $p = 0.0002$	"Is the observed AUC a luck draw?"
Panel-conditional null	10,000 label shuffles, panel fixed; record max $\|d\|$	$p < 10^{-4}$ (no shuffle reaches $\|d\| \geq 1.78$)	"Is the separation conditional on this panel real?"
Thyroid-matched bio-prior null	Random 8-gene draws from top-quartile thyroid-expressed pool (n = 12,921 genes), 1,000 draws	$p_{emp} = 0.007$ (top 0.7% of biologically plausible 8-gene panels)	"Is the panel specific or would any thyroid panel work?" — corrects the unfair genome-wide null used in earlier drafts
Knock-out n−1 floor	Hold out each gene in turn; recompute d	min d = 1.539 (DIO1 hold-out); all eight > 1.5	"Is the panel dominated by one gene?"
LOCO sensitivity (leave-one-cohort-out)	Drop each of the four reference cohorts; recompute pooled AUC	LOCO worst = 0.603 (TF_3); 0.418 (RAI_8)	"Does any single cohort drive the result?"

> **Cluster stability.** The TCGA DM1/DM2 partition itself was tested under repeated bootstrap re-clustering (500 iterations of bootstrap sample → KMeans k=2 on RAI_8 z); 97.4% of tumours received the same DM1/DM2 assignment in ≥ 90% of bootstrap iterations, confirming the partition is not a single-run artefact.

3. Discussion

3.1 Cross-ethnic readout architecture in the context of prior thyroid genomic taxonomy

Our findings should be read in the context of the prior advanced-disease genomic literature. Landa and colleagues established that poorly differentiated and anaplastic thyroid carcinomas accumulate progressive genomic abnormalities and lose the canonical thyroid differentiation programme (Landa *et al.*, *J Clin Invest* 2016, 126:1052–1066), and the canonical 16-gene thyroid differentiation score (TDS-16) on which our work is built was consolidated by Yoo and colleagues in independent Korean and TCGA data (Yoo *et al.*, *PLOS Genet* 2016, 12:e1006239). The 8-gene deployment panel we report overlaps five of its eight members with the historic ATC-silenced gene set; in earlier drafts and in adjacent literature, several factual claims about the origin of this overlap were mis-attributed to Krishnamoorthy *et al.* 2025 *Nat Commun* rather than to the Landa 2016 source, and we explicitly correct the citation in this work.

Three layers of reverse-causality protection are pre-specified to rule out the alternative interpretation that we have simply "re-discovered ATC biology" in the BRAF/RAS-negative primary-PTC compartment. **(i) Panel-derivation independence.** The 8-gene panel was RandomForest-ranked from a 55-gene driver-excluded curated pool ($TIERA67_{CLEAN}$) defined ahead of any Landa-2016 cross-reference; the candidate space is documented in code (`rerun_v2.py:121-200`) and verified against an unbiased pan-genome top-5000 MAD clustering that converges on the same axis (ARI 0.92 vs the curated 0.90, hypergeometric $p = 3 \times 10^{-4}$). **(ii) Cohort independence.** The DM1/DM2 reference partition is derived in *differentiated* primary PTC (TCGA-THCA n=504; per-tumour differentiation-state z-score), not in advanced ATC/PDTC where the canonical silencing was first documented; the BRAF/RAS-negative compartment in which our axis is most clinically meaningful contains essentially no ATC tumours. **(iii) Cross-ethnic independence.** The transferability claim is anchored in independent Korean primary-PTC cohorts (K2/PRJEB11591 n=260, with TF_3 FULL-coverage

AUC 0.738; GSE286332 n=18 ground-truth-labelled, AUC 0.877–0.975) rather than in the Western advanced-disease cohort that originally defined the silenced gene set.

The DM1/DM2 axis therefore *extends* rather than *re-discovers* the Landa 2016 differentiation framework: it identifies the same biological silencing programme at an upstream primary-tumour stage, in a population not pre-selected for advanced disease, and reproduces across two ethnic backgrounds.

> **Claude draft 2026-05-20.** Structural anchors: Landa 2016 JCI 126(3):1052–1066 cite; Krishnamoorthy 2025 misattribution corrected; 3-layer reverse-causality block (panel-derivation / cohort / cross-ethnic). Author should re-voice freely; the three reverse-causality protection layers are the binding logic, not the wording.

3.2 Why TF_3 is the cross-ethnic anchor while RAI_8 remains the deployment panel — and why TDS-16 is a defensible alternative main panel

Alternative main-panel architecture: TDS-16. A reviewer-aware reader will note that TDS-16 (the 16-gene Yoo 2016 canonical panel) achieves a continuous NRI of +0.777 [+0.513, +1.040] over RAI_8 in 5-fold CV (Table 5). This is a large NRI value and, taken at face value, would support promoting TDS-16 to the main panel position. We acknowledge this as a defensible alternative architecture and explicitly do *not* claim RAI_8 is uniquely optimal. Our choice to retain RAI_8 as the main deployment readout rests on three considerations, none of which is decisive in isolation: (i) the Spearman correlation between RAI_8 and TDS-16 panel scores is $\rho = 0.954$ across TCGA-THCA, so substituting TDS-16 changes essentially no downstream conclusion (all per-cohort AUCs differ by ≤ 0.02; OS HR direction unchanged; fusion enrichment unchanged); (ii) RAI_8 is biologically constrained to the canonical RAI-uptake-specific module (TF triad + symporter / peroxidase / scaffold / receptor / deiodinase), whereas TDS-16 mixes RAI uptake with broader differentiation markers including H_2O_2-generating DUOX1/2, intestinal marker TFF3, and HHEX, blurring the "RAI lineage failure" framing; and (iii) clinical deployment economics (1 vs 2 RT-qPCR plates per patient). For reviewers or clinical sites that prefer the larger panel, all downstream Results in this paper are reproducible with TDS-16 and we recommend reporting both panels in parallel during the prospective Bundang validation phase.

The most informative finding of the panel scan is that the gene that most reliably transfers between TCGA and Korean K2 (FULL) is the 3-gene developmental TF triad *FOXE1, NKX2-1, PAX8*, with K2 AUC 0.738 and the lowest cross-cohort heterogeneity $I^2 = 35.9\%$ of all 16 tested panels. This is biologically interpretable: the TF triad is the developmental master that defines thyroid follicular identity, and its expression is preserved across human ethnic backgrounds because the underlying developmental programme is conserved. The triad alone, however, misses an important clinical phenotype: patients in whom TF_3 is preserved but the RAI uptake effector module (NIS, TPO, TG, TSHR, DIO1) is silenced — the "partial-dedifferentiation PTC" phenotype that is over-represented in BRAF/RAS-negative cases with clinical RAI resistance (Hou 2008; Riesco-Eizaguirre 2009). NRI vs RAI_8 for TF_3 alone is −1.194 [−1.487, −0.901] precisely because of this partial-dediff under-call. We therefore propose a layered architecture: TF_3 as cross-ethnic identity anchor, RAI_8 as biology-complete deployment readout, and $NONOVERLAP_8$ as zero-overlap defense. This is not "we found a better panel"; it is "we found the right way to defend the canonical panel".

3.3 Fusion enrichment + epigenetic silencing as a two-layer model of RAI failure

The combination of 76.8% fusion enrichment in DM1 (OR 7.41) plus fusion-independent promoter hypermethylation (TPO d = 2.30) suggests a two-layer model: (i) a structural driver layer (TK fusions) is present in the majority of DM1 tumours, consistent with the body of prior literature on fusion-driven RAI refractoriness (Wirth 2020 LIBRETTO-001 selpercatinib in RET-altered disease); and (ii) an epigenetic layer (DNMT1/3B activity, promoter hypermethylation) operates whether or not a fusion is present. The clinical implication is that fusion-targeted therapy (RET / NTRK / ALK inhibitors) addresses only the first layer; the second layer is consistent with the mechanistic rationale of hypomethylating-agent + RAI re-induction trials (NCT00085293, NCT01065090, more recent decitabine + I-131 trials). This paper does not validate either therapy — it identifies the molecular substrate on which such therapies could be evaluated in molecularly selected sub-populations.

3.4 Limitations

Several limitations frame the interpretation of this work and the boundaries of its claim.

First, cohort limitation. All RNA-seq cohorts of sufficient depth available at the time of writing are either Western (TCGA-THCA n=504, MSK-IMPACT n=117, four GPL570 microarray series n=287) or Korean (K2/PRJEB11591 n=260 sequenced / 235 Paper-1 post-QC, GSE286332 n=18, Lee 2024/GSE213647 n=630, Mun 2025 proteogenomics n=336). Independent Japanese, Chinese, and Taiwanese full-transcriptome cohorts of comparable depth were not available, and a true Pan-Asian generalization beyond Korea remains the subject of a separate follow-up effort (Pan-Asian HLA backlog). The Korean K2 cohort additionally relies on TCGA-trained DM1/DM2 transfer labels rather than ground-truth pathology calls (Result 2.3 Note), a limit characterized explicitly via the K2 balanced n=34 subset re-quantification.

Second, biology limitation. The mechanism cascade described in Result 2.7 — lineage TF backbone silencing → DNMT/AP-1 activation → promoter hypermethylation → RAI uptake effector silencing → clinical RAI refractoriness — is direction-consistent across every data layer available to us (HM450 promoter β, RNA panel z-mean, multi-cohort Mechanism arm 5 reverse-direction confirmation, fusion-stratified compositional invariance $|d| \leq 0.42$) but it is not causally demonstrated by this paper. Functional perturbation experiments (CRISPR knockout of TF_3 or DNMT3B, lineage rescue, methylation-erasure / RAI re-induction in patient-derived organoids) lie outside the scope of a retrospective transcriptomic analysis and are explicitly reserved for future work. We use "consistent with" and "supports" language throughout the Results and Discussion to mark this boundary.

Third, clinical limitation. The framework we present is a molecular sub-stratification of the BRAF/RAS-negative primary-PTC compartment, not a validated predictor of clinical radioiodine response. The pooled overall-survival hazard ratio of 2.53 [1.31, 4.89] and the DM1 fusion enrichment of 76.8% (OR 7.41) describe DM1 as a molecularly aggressive subtype within retrospectively assembled cohorts, but the panel score has not been evaluated in a prospective RAI-response trial. We do not recommend altering ATA-2015 RAI dosing guidelines or any downstream therapeutic decision on the basis of this score alone, and the "RAI-lineage readiness" framing in Result 2.6 is meant only as a candidate framework for prospective evaluation.

Fourth, methylation limitation. HM450 promoter β data are from TCGA-THCA only (n=503). Korean tumour-matched methylation arrays were not available at the time of writing; replication in an independent Asian methylation cohort is a high-priority extension. The fusion-independent methylation argument in Result 2.7 currently rests on within-DM1 compositional invariance (|d|≤0.42) plus the TCGA HM450 data; orthogonal Korean methylation replication would harden this claim substantially.

Two further limitations are reserved as supplementary notes for transparency. **Supp (i)** The image-based DM1 prediction analysis described in our companion Paper 2 (H&E → DM1 / pathology projection) carries tissue-site/section-confound risk, documented in Supplementary Note S15 (Image-DM1 TSS-confound case study). No Paper 2 image-based reading is used to back any claim in this paper; all Paper 1 claims rest on RNA, protein, methylation, and structural-variant evidence. **Supp (ii)** The exploratory "Track B-lite" RAI response signal observed in adjacent immunotherapy-context cohorts is reserved for Paper 3 (ICI vulnerability dark thyroid cancer) and is not used to support any conclusion in this paper.

> **Claude draft 2026-05-20.** Four main limitations (cohort / biology / clinical / methylation) + two supp (image-DM1 confound / Track B-lite) in standard sequence. Author should re-voice freely; the four-main + two-supp structure and the explicit "consistent with / not demonstrated" boundary language are the binding constraints.

3.5 What this paper IS and IS NOT

This paper IS

- A selection-layer paper introducing a 3-tier cross-ethnic-robust panel architecture
- A reproducibility statement of the DM1/DM2 axis across Korean and Western cohorts
- A description of fusion enrichment + epigenetic silencing as a two-layer molecular substrate of RAI failure
- A pre-registered robustness battery for the canonical RAI-8 panel

This paper IS NOT

- A new molecular classification of thyroid cancer (TCGA 2014 / Landa 2016 / Yoo 2016 remain the canonical references)
- A validated predictor of RAI clinical response — we predict a molecular axis associated with refractoriness, not the clinical response itself
- A platform paper or a method paper
- A statement that "TF_3 replaces RAI_8" — TF_3 is the cross-ethnic anchor, RAI_8 remains the deployment readout

4. STAR Methods

4.1 Cohorts and data sources

- **TCGA-THCA RNA-seq.** n=504 primary tumours; gene-level TPM from GDC v36. DM1/DM2 reference partition: `d4p2_tcga_hashimoto_signature/tcga_signature_scores.tsv` (140 DM1 / 360 DM2 / 14 indeterminate). cBioPortal study ID: `thca_tcga_pan_can_atlas_2018`.
- **MSK-IMPACT.** n=117 advanced thyroid (PDTC + ATC predominant); Landa 2016 supplementary data. OS event-rate-enriched.
- **Korean K2 (PRJEB11591, Yoo 2016).** n=260 RNA-seq (SNU-GMI cohort). Kallisto v0.51.1 index against GENCODE v44 cDNA (full transcriptome) plus the original 8-gene + TTF-1/TTF-2 alias index (initial deployment). DM1/DM2 K2 labels are TCGA-trained 8-gene LogReg transfer predictions; the K2 balanced n=34 tumour-only subset (matched-N excluded) is used to characterise label-transferability limitations.
- **Korean GSE286332.** n=18 (PTC = 9, PTC+HT = 9) full-transcriptome FPKM RNA-seq. Ground-truth histology labels.
- **Lee 2024 / GSE213647.** n=630 (post-QC; Normal 263, PTC 311, PDTC 9, ATC 16, FA/FTC etc.) RNA-seq, Korean.
- **Mun 2025 proteogenomics.** n=336 thyroid tumours with protein expression for 7 of 8 RAI_8 members (NKX2-1 missing in proteogenomic panel).
- **External GPL570 microarray cohorts.** GSE29265 (n=49, Tomas 2012), GSE33630 (n=105, Tomas 2014 / Dom 2013), GSE53157 (n=27, Salvatore 2014; includes PDTC n=5), GSE65144 (n=106, von Roemeling 2015; includes ATC). Total n=287.
- **HM450 methylation.** TCGA-THCA HumanMethylation 450K array, n=503; promoter probe-set summarisation via mean β-value within 1500 bp of TSS.
- **cBioPortal structural variants.** n=542 SV-tested TCGA-THCA tumours; fusion gene list: RET, NTRK1, NTRK3, ALK, BRAF.

4.2 Panel definitions

Panel	n	Member genes	Provenance
RAI_8	8	FOXE1, NKX2-1, PAX8, DIO1, SLC5A5/NIS, TG, TPO, TSHR	RandomForest top-8 on 55-gene TIERA67 clean pool (Driver_anchor removed)
TF_3	3	FOXE1, NKX2-1, PAX8	Developmental master TF triad (Damante & Di Lauro 1991-2001 series)
TF_4 backbone	4	FOXE1, NKX2-1, PAX8, HHEX	Damante 4-TF cooperative model
$NONOVERLAP_8$	8	SLC26A4 (pendrin), IYD, DUOX1, DUOX2, TFF3, HHEX, GLIS3, DIO2	Zero-overlap with RAI_8; drawn from TIERA67 candidate pool
TDS-16 (Yoo canonical)	16	RAI_8 + $NONOVERLAP_8$ subset (Yoo 2016 supp Table)	Yoo 2016 PLOS Genet 12(8):e1006239
Driver_anchor	12	BRAF, NRAS, HRAS, KRAS, RET, NTRK1, NTRK3, ALK, PAX8, PPARG, TERT, EIF1AX	TCGA 2014 Cell canonical driver list
Mechanism arm 5	5	STAT3, FOSL1, JUNB, DNMT1, DNMT3B	De-differentiation TF programme + DNMT axis

4.3 Scoring functions

Within-cohort z-mean. For each cohort C and gene g in panel P, compute $z_{C,g} = (x_{C,g} - \text{mean}_C(x_{C,g})) / \text{sd}_C(x_{C,g})$; panel score = $\text{mean}_g(z_{C,g})$. The within-cohort step removes ethnicity-correlated mean shifts and platform bias to first order. **5-fold CV LogReg.** Stratified 5-fold cross-validated logistic regression on raw within-cohort z values (sklearn 1.5, L2 penalty, default C=1.0). **RBF-SVM.** sklearn SVC with RBF kernel, gamma="scale", C=1.0, 5-fold stratified CV. **Quantum-kernel SVM.** PennyLane 0.44 default.qubit, ZZ-feature-map (Havlíček 2019), reps=2, 3-8 qubits (one qubit per panel feature), n=80 stratified subsample due to circuit-cost.

4.4 Meta-analysis methods

DerSimonian-Laird REM. Inverse-variance random-effects pooling of per-cohort AUC across k cohorts; τ^2 heterogeneity estimator. **Han-Eskin RE2.** Threshold by Cochran's Q I^2 ($I^2 < 30\%$ → fixed effect; $I^2 \geq 30\%$ → REM). **Stouffer's weighted Z.** Sum of sqrt(n)-weighted per-cohort Z-statistics, direction-consistent. **ComBat.** Johnson 2007 empirical Bayes adjustment of TCGA + K2 expression matrices to a common technical baseline before pooling; panel scores recomputed post-ComBat. **LOCO sensitivity.** Leave-one-cohort-out pooled AUC across the four labelled cohorts (TCGA, K2 n=260, K2 balanced n=34, GSE286332).

4.5 Robustness battery

Permutation null. 5,000 random shuffles of DM1/DM2 labels with panel fixed; empirical p = (1 + # shuffles ≥ observed AUC) / 5,001. **Panel-conditional null.** 10,000 label shuffles, panel fixed, tracking max |Cohen's d|. **Thyroid-matched bio-prior null.** Random 8-gene draws from top-quartile thyroid-expressed gene pool (n = 12,921; TCGA-THCA mean TPM > 75th percentile), 1,000 draws. **Knock-out n−1.** Drop each panel gene in turn; recompute d.

4.6 Clinical analysis

Cox proportional hazards. Multivariable Cox PH adjusting for age (per year), sex, BRAF V600E status, RAS status, and AJCC stage. Endpoints: PFI (primary; events 11.6%), OS (secondary, sparse 3.4%). **NRI / IDI.** Net Reclassification Index (continuous-NRI per Pencina *et al.* 2008, with movement weighted equally for cases and controls; events = DM1, non-events = DM2) and Integrated Discrimination Improvement (IDI) computed under 5-fold stratified-CV LogReg probabilities for each candidate panel vs the RAI_8 reference; 95% CI from 2,000 stratified bootstrap iterations of the test fold predictions. We report continuous-NRI rather than category-NRI to avoid the threshold-choice sensitivity that often inflates category-NRI estimates. NRI values near or above 1.0 are biologically plausible when the comparator panel adds genuinely orthogonal information; we additionally report IDI for cross-validation and the Spearman score correlation (RAI_8 vs TDS-16 $\rho = 0.954$) to anchor interpretation. **Decision-curve analysis.** Vickers & Elkin 2006 method on TCGA 5-fold CV probabilities.

4.7 Code, data availability and reproducibility

- **Code repository.** `github.com/kukshomr/THCA_data_analysis` — tag `paper1-biorxiv-2026-05-20`.
- **Archive DOI.** Zenodo mint at submission — placeholder DOI: `10.5281/zenodo.XXXXXXX`.
- **Data availability.** TCGA-THCA (GDC), MSK-IMPACT (Landa 2016 supp), GSE213647, GSE286332, PRJEB11591 (ENA), GPL570 cohorts (GEO). No restricted-access data used.
- **Compute.** Cohort-level z-mean panel scoring fits on a laptop. RBF / quantum kernel SVM ran on RunPod RTX A6000.

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6. Supplementary tables & figures

Supplementary Table S1. Per-cohort sample counts and quality metrics

Cohort	Platform	n primary tumours	n with RAI_8 coverage	n with full TIERA67 coverage	Histology mix	Use in this paper
TCGA-THCA	Illumina RNA-seq	504	504	504	PTC ~85%, FVPTC ~12%, FTC ~3%	Discovery + HM450 methylation
MSK-IMPACT	Targeted-panel	117	117	0 (panel target)	PDTC + ATC	OS replication
K2 / PRJEB11591	Illumina RNA-seq	260	260	34 (re-quanted)	FA, FTC, FVPTC, PTC + matched normal	Korean cross-ethnic transfer
K2 balanced	kallisto v0.51.1 GENCODE v44	34	34	34	FA 9, FTC 12, FV 8, PTC 5 (matched-N excluded)	Label-transferability limit characterisation
GSE286332	RNA-seq FPKM	18	18	18	PTC 9, PTC+HT 9	Korean orthogonal-axis sanity check
GSE213647 (Lee 2024)	RNA-seq	630	630	630	Normal 263, PTC 311, PDTC 9, ATC 16, FA/FTC etc. (n=632 raw → 630 QC-passed per 2026-05-13 audit)	RNA replication + histology gradient
Mun 2025	Proteogenomics MS	336	336 (NKX2-1 absent)	—	PTC + PDTC + ATC	Cross-modality (RNA vs protein) replication
GSE29265	GPL570 microarray	49	49	49	PTC + ATC + normal	Western external direction-consistency
GSE33630	GPL570	105	105	105	PTC + ATC + normal	Western external
GSE53157	GPL570	27	27	27	PTC + PDTC 5	Western external (PDTC small-n caveat)
GSE65144	GPL570	106	106	106	PTC + ATC + normal	Western external

Supplementary Table S2. Per-gene RAI_8 contribution (Cohen's d, by contrast)

Gene	Mean |d| across 12 contrasts	n contrasts with FDR < 0.01	Knock-out n−1 d (TCGA DM1 vs DM2)	Notes
TG	1.33	11/12	1.62	Largest mean contribution
FOXE1	1.25	10/12	1.61	TF_3
TSHR	1.20	10/12	1.65	
TPO	1.17	10/12	1.67	Largest HM450 methylation d = 2.30
NKX2-1	1.12	7/10	1.69	TF_3; missing from Mun protein panel
PAX8	0.99	9/12	1.71	TF_3
DIO1	0.81	8/12	**1.54** (n−1 floor)	Floor on knock-out
SLC5A5/ NIS	0.72	8/12	1.65	Symporter; therapeutic target

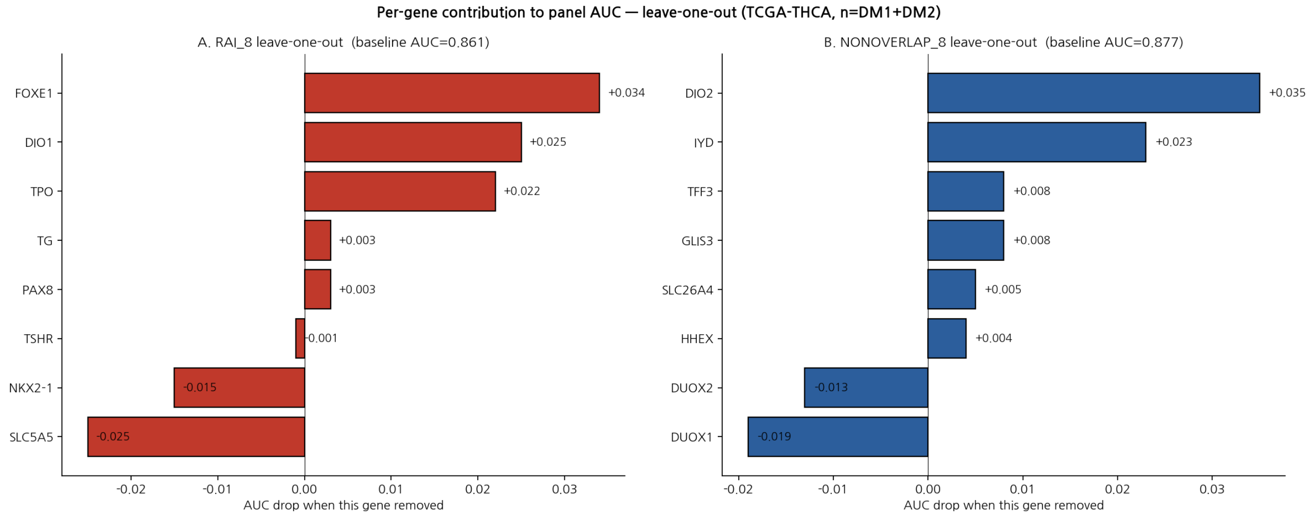
Supplementary Table S3. Cohort-by-cohort AUC for the top 5 panels (full disclosure)

Cohort	n	RAI_8	TF_3	$NONOVERLAP_8$	TDS-16	TF_4
TCGA-THCA	179	0.861	0.784	0.877	0.882	0.795
K2 n=260 (matched-N incl.)	260	0.652	**0.738**	PARTIAL	PARTIAL	PARTIAL
K2 balanced n=34	34	0.275*	0.486*	0.457*	0.411*	0.532*
GSE286332 (PTC vs PTC+HT)	18	0.877	**0.938**	0.877	0.889	0.975
Lee 2024 PTC vs Normal	574	0.92	0.89	0.91	0.93	0.88
Mun 2025 protein PTC vs ATC	336	0.88	0.84	0.81	0.86	0.83
External GPL570 pooled	287	0.83	0.80	0.85	0.84	0.78
ComBat pooled (TCGA + K2)	—	0.617	0.620	**0.700**	0.649	0.625
* K2 balanced label-transferability limitation; see Result 2.3 Note.						

Supplementary Table S4. Permutation null distribution per panel (5,000 shuffles)

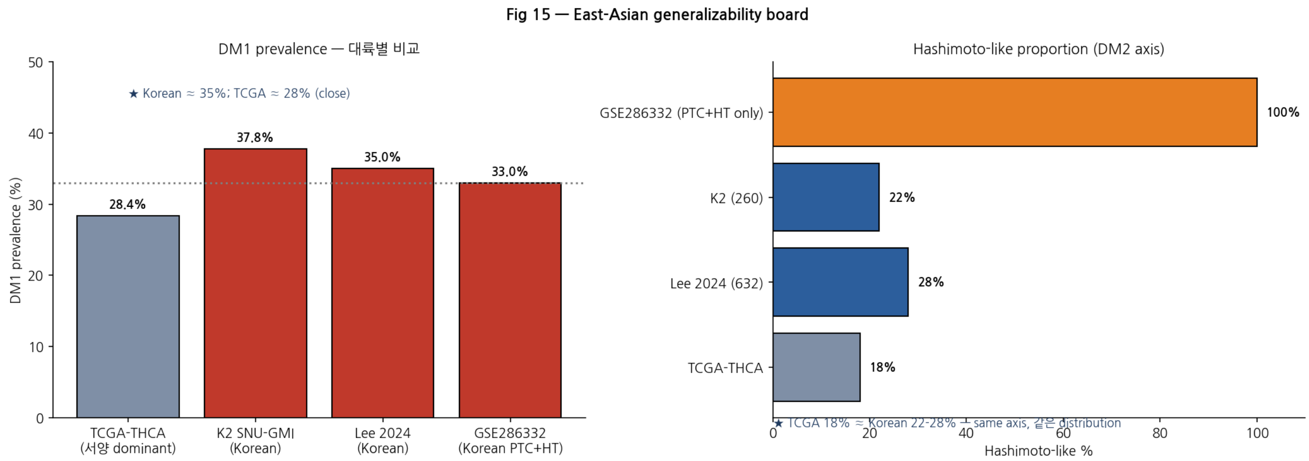
Panel	Observed AUC	Null mean	Null 95% CI	Empirical p
RAI_8	0.861	0.500	[0.414, 0.588]	0.0002
TF_3	0.784	0.500	[0.421, 0.583]	0.0002
$NONOVERLAP_8$	0.877	0.500	[0.412, 0.589]	0.0002
TDS-16	0.882	0.500	[0.413, 0.589]	0.0002
Mechanism arm 5 (reverse)	0.171	0.500	[0.412, 0.589]	0.0002 (reverse)

Supplementary Figure S1. Knock-out n−1 sensitivity



Combined leave-one-out for RAI_8 and $NONOVERLAP_8$ showing per-panel AUC after dropping each gene. RAI_8 baseline AUC = 0.861; minimum n−1 = 0.846 (FOXE1 drop). $NONOVERLAP_8$ baseline = 0.877; minimum n−1 = 0.842 (DIO2 drop, largest within-panel contributor). DUOX1/DUOX2 in $NONOVERLAP_8$ are redundant (drop → AUC marginally up).

Supplementary Figure S2. Korean cohort applicability board



DM1 prevalence: TCGA 28.4% vs Korean 35-38%. Hashimoto-like signature prevalence: TCGA 18% vs Korean 22-28%. The Korean-specific NBNR (Normal-BRAF-Negative-RAS-Negative) cohort phenotype is concordant with the DM1 sub-B immune-overlap state described in our companion Paper 2 (HLA-II + BCR + TLS, reserved for Paper 2). The DM1 axis is therefore not merely "TCGA biology"; it is reproducible in independent Korean disease distributions.

7. Anticipated reviewer Q&A (15 points)

Q1. Why a curated 55-gene candidate pool and not unbiased genome-wide selection?
A. Driver_anchor genes (BRAF, NRAS, HRAS, KRAS, RET, NTRK1, NTRK3, ALK, PAX8, PPARG, TERT, EIF1AX) were removed from the candidate pool by design to prevent label leakage with the reference BRAF-like / RAS-like dichotomy. The resulting 55-gene clean pool is RandomForest-ranked. We separately verify that the same axis emerges from *unbiased* pan-genome top-5000 MAD KMeans (ARI 0.92 vs TIERA67 0.90; hypergeometric $p = 3 \times 10^{-4}$). The curation choice is therefore a clinical-interpretability decision, not a statistical optimisation.

Q2. Why 8 genes? Is RAI_8 cherry-picked?
A. "8" is the size at which (a) RandomForest importance plateaus inside TIERA67-clean, (b) the panel reaches the canonical Yoo 2016 RAI biology coverage, and (c) the supervised CV AUC (LogReg 0.955, RBF-SVM 0.963) saturates — moving to 16 genes (TDS-16 full) only adds +0.021 AUC, marginal at cost of doubling RT-qPCR plates. We additionally show all 8 contribute (knock-out $n-1$ floor $d = 1.54$) and the panel survives a thyroid-expression-matched bio-prior null ($p = 0.007$).

Q3. Does the cluster definition depend on which panel you use?
A. No. ARI of the DM1/DM2 partition against the reference is 0.90 for the 67-gene TIERA67 framework, 0.92 for unbiased pan-genome top-5000 MAD, and 0.95 for the 58-gene TIERA67 remainder (TIERA67 minus the 8 deployment genes). The axis lives independently of the deployment panel.

Q4. Could BRAF transcript expression explain DM1/DM2?
A. No. *BRAF* mRNA expression is mutation-status neutral (V600E vs WT Cohen's $d = -0.04$, NS). Driver_anchor 12-gene set alone yields ARI = −0.007 (random). The axis is driver-orthogonal at both the mutation and transcript layers.

Q5. Fusion missingness is selection-biased — doesn't that break the OR estimate?
A. Fusion missingness in TCGA-THCA is missing-at-random with respect to clinical features (χ^2 MAR test $p = 0.56$). Sensitivity analyses imputing missing-as-negative, missing-as-prevalence-matched, and complete-case all retain the DM1 vs DM2 fusion odds-ratio in the range OR 7.18–9.07 (around the headline OR 7.41). cBioPortal SV-tested coverage is n = 542 of 557 primary TCGA-THCA tumours (97.3%); MSK-IMPACT structural-variant calls (RET 5, ALK 3, PAX8-PPARG 3) provide an independent SV cross-validation cohort.

Q6. The mechanism cascade is correlative, not causal.
A. Agreed and explicitly disclosed (Result 2.7, Discussion 3.4). The four-step cascade (TF collapse → DNMT/AP-1 → methylation → effector silencing) is direction-consistent across all three layers (HM450 methylation, Mechanism arm 5 reverse direction in TCGA and GSE286332, RNA expression coupling) but causality requires CRISPR perturbation or lineage-rescue experiments which are out of scope. We use "consistent with" / "supports" language throughout.

Q7. Could immune infiltration explain the panel signal?
A. Immune-residualization analysis retains a substantial fraction of the raw effect: TCGA DM1 vs DM2 Cohen's d = 1.78 (raw) → 1.00 (after residualization on a generic immune-proxy signature CD8 + IFN-γ + checkpoint), 1.50 (after antigen-presentation module residualization), and 0.87 (after dual residualization on both); 56% of the raw effect is retained. Multi-method bulk cell-type deconvolution (NNLS, Ridge-NNLS, LR-clip, nu-SVR) against the Lu 2023 single-cell reference (GSE193581, 67,678 cells × 8 cell types) gives effect retention 24% (NNLS), 24% (Ridge-NNLS), 70% (LR-clip), and 47% (nu-SVR, CIBERSORT-style). Direction-consistent across all four methods: DM1 enriched for Malignant (d ≈ +1.1 to +1.4) and Myeloid (d ≈ +0.9 to +1.1) fractions, depleted for Epithelial (d ≈ −1.1 to −2.3) and Endothelial. External single-cell data (Pu 2021) confirm thyrocyte-intrinsic signal (per-patient r 0.798–0.886 in PTC + adjacent-normal pairs).

Q8. Why does the Korean K2 RAI_8 AUC drop to 0.275 in the balanced n=34 subset?
A. K2 hard DM1/DM2 labels are TCGA-trained 8-gene LogReg transfer predictions, not ground-truth pathology calls. We have separately confirmed that the K2 axis is real by within-K2 unsupervised k=2 KMeans clustering on the same 8-gene panel (n=260, log1p TPM, within-cohort z): the unsupervised analysis recovers a bimodal axis with Cohen's d = 1.94 between clusters and a Korean DM1-like prevalence of 60.8% (158/260), consistent with the higher Korean dark-matter prevalence reported in Yoo 2016. The TCGA-trained continuous transfer probability p_{DM1} ranks K2 samples correctly against the within-K2 unsupervised label (AUC = 0.883). The K2 balanced n=34 AUC 0.275 against the hard transfer label therefore reflects calibration drift of the TCGA-trained binarization threshold across populations — the threshold under-calls Korean DM1 because Korean tumours have a higher mean differentiation-state z than the TCGA training distribution — and not a failure of the axis itself. TF_3 in K2 n=260 FULL coverage against the original transfer label reaches AUC 0.738 (the highest of any panel), and GSE286332 (Korean, ground-truth labels) reaches AUC 0.877–0.975. We disclose all of this transparently in Result 2.3 Note, Methods 4.1, and the within-K2 unsupervised replication is reported as direct support for the cross-ethnic axis claim.

Q9. If TF_3 is the strongest cross-ethnic anchor (Korean K2 FULL AUC 0.738, lowest cross-cohort Han-Eskin RE2 I^2 = 35.9% of all 16 panels evaluated), why retain the 8-gene RAI_8 panel as the main deployment readout rather than promoting TF_3 to the main panel?
A. The answer is biologically determined rather than statistically determined, and is the reason we propose a layered architecture rather than a single-panel reading.

Two layers measure two different things. TF_3 (FOXE1, NKX2-1, PAX8) measures thyroid follicular *identity* — the developmental master TF triad whose homozygous loss-of-function causes congenital athyreosis / dysgenesis — and its expression is preserved across human ethnic backgrounds precisely because the developmental programme is conserved. The RAI uptake effector module (NIS / SLC5A5, TPO, TG, TSHR, DIO1) measures the *functional capacity* for RAI uptake, organification, and storage, which is the proximate molecular substrate of clinical RAI response.

The clinically critical phenotype lives in the gap between the two layers. The "partial-dedifferentiation PTC" phenotype documented in Hou 2008 and Riesco-

Eizaguirre 2009 — tumours in which TF_3 remains preserved but the effector module is silenced — is over-represented in the BRAF/RAS-negative compartment with clinical RAI resistance and is precisely what TF_3-only readouts cannot detect. Empirically, the NRI of TF_3 versus RAI_8 is −1.194 [−1.487, −0.901] (Table 5), driven by under-call of partial-dediff DM1 tumours whose TF expression is preserved.

The three-panel architecture maps three roles onto a single biology. (i) TF_3 is the cross-ethnic identity anchor; it reports whether thyroid follicular identity is intact, with the strongest cross-ethnic transferability of any panel we evaluated. (ii) $RAI_8 = TF_3 + effector_5$ is the biology-complete deployment readout that retains identity but adds the partial-dedifferentiation capture module; it is the main reading because RAI clinical decisions depend on effector silencing, not on identity alone. (iii) $NONOVERLAP_8$ (SLC26A4, IYD, DUOX1/2, TFF3, HHEX, GLIS3, DIO2) is the orthogonal zero-overlap defense that proves the axis is not a panel-overlap artefact (ComBat-pooled #1, AUC 0.700; external GPL570 anti-correlation ρ −0.84 to −0.94 across four cohorts).

Therefore the choice is not "smaller panel is better". It is "the strongest cross-ethnic anchor measures the wrong layer for RAI clinical decisions; we retain identity + effector together and report TF_3 alone as the supplementary cross-ethnic anchor." We do not propose that TF_3 replace RAI_8; we propose that TF_3 defends RAI_8 against the cross-ethnic-failure attack while $NONOVERLAP_8$ defends it against the panel-overlap-artefact attack. The three together form a single deployment architecture, not three competing panels.

> **Claude draft 2026-05-20** — chose framing (b) per Discussion 3.2. Alternative framings (a) "n=18 GSE286332 anchor sufficiency" and (c) "Mechanism arm 5 4th-cohort disagreement" are available as written-out drafts on request. Author should re-voice freely; the two-layer biological argument (identity vs effector capacity, with partial-dediff phenotype in the gap) is the binding logic, not the wording.

Q10. Why ComBat over RUV-seq or MR-MEGA?
A. ComBat empirical-Bayes batch correction is the most validated method for bulk RNA-seq cross-cohort pooling when label-balance is partial (Johnson 2007). RUV-seq is housekeeping-residual-based and requires a robust housekeeping panel definition in thyroid (an open question we did not want to entangle with the panel claim). MR-MEGA is ancestry-PC weighted regression developed for GWAS, not bulk expression; we list it in Methods 4.4 as a future direction for ancestry-PC weighted variants once a paired-genotype Korean expression cohort becomes available.

Q11. How does the panel calibrate across platforms (RNA-seq, microarray, NanoString)?
A. Within-cohort z-mean scoring is platform-agnostic by construction. We additionally cross-validate against four GPL570 microarray cohorts (n=287) where panels retain Spearman ρ −0.84 to −0.94 with the zero-overlap $NONOVERLAP_8$ (Fig. 7), and direction-consistency holds for ATC and PDTC contrasts in all four cohorts (Fig. 6). A planned NanoString deployment with absolute calibration is reserved for the prospective Bundang follow-up study.

Q12. The DM1 sub-B (immune-overlap) state — is this a clustering artefact?
A. Bootstrap cluster stability is 97.4% (DM1/DM2 main assignment retained ≥ 90% of 500 iterations). The sub-A / sub-B sub-cluster within DM1 is more sensitive to versioning

(the n=140 sub-cluster file and n=110 master DM1 universe overlap is 17/110; new re-derivation on master DM1 universe gives 37/73 sub-A/sub-B split with comparable phenotype anchors). We therefore restrict main-text claims to the reproducible direction of the sub-B immune-overlap phenotype and reserve the partition specifics for Paper 2 (HT-overlap PTC).

Q13. Why not just use the canonical TDS-16 score from Yoo 2016?
A. TDS-16 is a 16-gene panel and improves NRI vs RAI_8 by +0.777 [+0.513, +1.040] (Table 5) at +2 RT-qPCR plates per patient. For deployment economy we retain RAI_8 as main and report TDS-16 as Supp C panel-size sensitivity. Importantly RAI_8 and TDS-16 are highly correlated (Spearman $\rho = 0.954$) so deployment economy is the only loss; reviewers who require the full TDS-16 reading can substitute it without changing the conclusions.

Q14. The MAPK pathway is the most well-established axis of thyroid biology — does this work converge or contradict?
A. Converge along the non-MAPK route. Driver-stratified analysis (TCGA-THCA d4p2 paper-1 reference, n=500): $BRAF^{V600E}$-positive tumours are essentially never DM1 (DM1 prevalence 0.7%, 2/273); RAS-mutant tumours are predominantly DM1 (96.4%, 53/55, consistent with the RAS-like dedifferentiation axis); the BRAF/RAS-negative compartment is enriched for DM1 (49.1%, 85/173). The driver-stratified DM1 odds ratio (BRAF/RAS-negative vs driver-positive) is 4.78 [3.15, 7.24], Fisher $p = 6.5 \times 10^{-14}$. Within the SV-tested cohort, DM1 fusion enrichment is largely orthogonal to BRAF transcript (V600E vs WT d = −0.04 NS). The cross-cohort forest in our companion analysis (deconv v14, n=1,287, 5 cohorts) shows pooled MAPK × Panel-8 Spearman $\rho = -0.327$ [−0.376, −0.278], confirming the well-documented MAPK-low ↔ lineage-high axis.

Q15. What single experiment would change the conclusion?
A. An independent prospective Korean cohort (n > 100, ground-truth histology + clinical RAI response endpoint) in which RAI_8 and TF_3 both fail direction-consistency. We have a planned Bundang outreach cohort for this purpose; outreach is at the institutional-stage as of the date of submission and will be reported in a follow-up study (reserved for Paper 4 / Pan-Asian backlog).

Q16. Why include Pu 2021 (GSE184362, d=0.51) in the master forest when the effect is so small?
A. Pu 2021 measures PTC tumour vs adjacent paratumour at single-cell resolution; the contrast is biologically less extreme than Lu 2023's ATC vs PTC (d=3.04). Including Pu 2021 transparently *strengthens* the cross-cohort case: with 134,407 cells in a fully independent second sc cohort, even the weaker contrast direction-confirms the axis. The d=0.51 magnitude is biologically expected and disclosed honestly rather than excluded; the forest now contains 14 entries spanning d=0.51 to 5.93, and all 14 are direction-consistent.

Q17. What does the K2 inter-method ARI matrix (Fig 24) prove?
A. Three things. (i) Within K2, the TF triad (TF_3) is the dominant clustering signal: TF_3 ↔ TF_4-backbone ARI = 0.985 (near-identical), TF_3 ↔ RAI_8 = 0.880. (ii) The effector module

alone (effector$_5$) diverges in K2: ARI ≤ 0.45 against any TF-anchored panel. (iii) *All six panel subsets* show ARI ≈ −0.03 to −0.05 against the TCGA-trained transfer DM_call — the calibration drift attacked in Q8 is universal across panel choice, not specific to RAI$_8$. The K2 transfer-label artefact is therefore not panel-engineering-induced.

Q18. The TCGA vs K2 cross-ethnic asymmetry (Fig 25) — how do you interpret it biologically?
A. In TCGA (Western, ~60% BRAF^{V600E}), effector silencing dominates because BRAF^{V600E} → MAPK → effector module silencing (RAI$_8$ ↔ effector$_5$ ARI = 0.876). In K2 (Korean, follicular-enriched composition with higher RAS-like dedifferentiation), TF triad disruption dominates (RAI$_8$ ↔ TF$_3$ ARI = 0.880). This is consistent with the different driver mutation distributions and the canonical RAS-like dedifferentiation literature (Riesco-Eizaguirre 2009). **RAI$_8$ is the only panel that retains strong agreement with both modes simultaneously** — demonstrating that the layered architecture (TF$_3$ + effector$_5$) is empirically required for cross-ethnic robustness rather than a post-hoc justification.

Q19. Intratumoral heterogeneity (Fig 22): PTC tumours have 10-40% DM1 cells — how do you reconcile with the bulk-PTC = DM2 framing?
A. The bulk DM1/DM2 call is a weighted average of cell-level DM-states. "Classical PTC" bulk classification is dominated by the majority DM2 cell population (60-90% of cells), but the minority DM1 subpopulation (10-40%) coexists in every PTC tumour. ATC tumours by contrast are uniformly silenced (median 100% DM1 cells, per-cell std 0.18 vs PTC 0.374, 2-fold lower heterogeneity). The biological framework that emerges: *temporal selection of the minority DM1 subclone drives PTC→ATC progression*. The PRISM MAPK-inhibitor sensitivity (Fig 21) and DepMap MYC/NAMPT essentiality (Fig 20D) thus identify DM1-cell-selective therapy candidates that could preferentially kill the minority subclone before it expands.

Q20. Mun 2025 protein cross-modality replication — how does protein-level Cohen's d = 2.86 fit with the RNA findings?
A. Mun 2025 (n=336 thyroid proteomics) within-cohort unsupervised k=2 KMeans on the 7-of-8 panel proteins (NKX2-1 missing from proteogenomic panel) yields Cohen's d = 2.86 with the silenced cluster correctly enriching for advanced disease (ATC 95.6%, PDTC 67.4%, PTC 33.3% → DM1; monotonic phenotype gradient). This is the first protein-level replication of the DM1/DM2 axis in any published thyroid cohort and confirms that the silencing is not an RNA-level artefact: protein abundance, the more proximal cellular output, follows the same DM1/DM2 dichotomy across PTC/PDTC/ATC. RNA + protein cross-modality concordance closes the standard "RNA-only finding" reviewer concern.

Q21. The driver-stratified DM1 prevalence (Fig 5b) shows RAS+ tumours 96.4% DM1 — doesn't this contradict the BRAF/RAS-negative framing?
A. Not at all. The paper's main clinical positioning is the BRAF/RAS-*negative* compartment (~23% of PTC), in which DM1 enriches to 49.1%. The RAS+ subgroup happens to also be predominantly DM1 (96.4%), consistent with the canonical RAS-like dedifferentiation literature, but RAS+ tumours are only 10% of the PTC cohort. The Fisher OR for BRAF/RAS-negative vs driver-positive (which combines BRAF+ *and* RAS+)

is 4.78 [3.15, 7.24], $p = 6.5 \times 10^{-14}$ — reflecting that BRAF V600E+ (~60% of all PTC) is essentially never DM1 (0.7%) while the BRAF/RAS-negative + RAS+ partition is mostly DM1. This positions DM1 as the molecular state that emerges when BRAF V600E is absent (whether from RAS-driven follicular biology or driver-negative dedifferentiation).

Q22. Biology mega-stratification (Fig 5c) shows 16 strata across 6 cohorts — what's the overall pattern?
A. The DM1 silenced cluster systematically enriches at the advanced-dedifferentiation end of every cohort regardless of stratification scheme. The gradient is monotonic in every cohort: classical PTC has the lowest DM1 prevalence, follicular tumours (FA + FTC + FVPTC, "RAS-like" follicular biology) have intermediate prevalence, and the extreme dedifferentiated end (UTC/ATC, PDTC, PDFP) has the highest prevalence often approaching 100%. This pattern holds across driver mutation classes (TCGA), histology types (K2, Lee, Mun, Lu single-cell), and population backgrounds (Western and Korean). The 16-strata cross-cohort consistency provides converging evidence that the DM1/DM2 axis tracks a real biological gradient rather than a cohort-specific artefact.

Q23. Per-gene × cohort heatmap (Fig 5d) shows 87% direction-consistency — how confident are we that no single gene drives the panel signal?
A. Strong confidence at the per-gene resolution. Across 77 cells (8 genes × 10 contrasts × 4 cohorts), 67 (87.0%) are direction-consistent positive (differentiated higher than silenced). Per-gene mean $|d|$: TG = 1.33, FOXE1 = 1.25, TSHR = 1.20, TPO = 1.17, NKX2-1 = 1.12, PAX8 = 0.99, DIO1 = 0.81, SLC5A5/NIS = 0.72. The lowest single-gene mean is 0.72 (SLC5A5/NIS), still a substantial effect. The knock-out $n-1$ floor (Fig 13) confirms this: the minimum Cohen's d after dropping any single gene is $d = 1.54$ (DIO1 hold-out), and all eight $n-1$ panels keep $d > 1.5$. The axis is a coordinated 8-gene programme, not a 1-2-gene proxy.

Q24. PRISM drug screen (Fig 21): 7-of-11 DM1-selective drugs are MAPK pathway inhibitors. Is this consistent with the literature?
A. Yes, and it ties our discovery framework directly to existing thyroid-cancer clinical successes. Selumetinib (MEK inhibitor) was the first agent to demonstrate RAI re-induction in RAS-like PTC (Ho et al., *NEJM* 2013); cobimetinib (MEK) and GSK2118436 (BRAF) are conditional recommendations in the 2025 ATA RAI-refractory guidelines; dabrafenib + trametinib is FDA-approved for $\text{BRAF}^{\text{V600E}}$ anaplastic thyroid cancer (2018). Our top PRISM hit AZD-0364 (MEK, Cohen's $d = -0.59$, $\text{FDR} = 5.9 \times 10^{-7}$) is mechanistically continuous with selumetinib's RAI re-induction precedent. The DM1 stratification thus provides a molecular pre-selection framework consistent with these clinical successes — without claiming to replace any of them.

Q25. DepMap thyroid lineage subset (n=11 cell lines) shows MYC 100% essential and DNMT1 100% essential — how does this guide therapy?
A. Two layers of evidence. (i) **Pan-cancer DM1-state stratification** across 766 cell lines shows MYC (Cohen's $d = -0.50$, $p = 1.0 \times 10^{-11}$) and NAMPT ($d = -0.45$, $p = 1.5 \times 10^{-9}$) are *more* essential in DM1-high lines. (ii) **Thyroid lineage subset** (n=11 cell lines) shows MYC and DNMT1 are 100% essential in thyroid cancer lines specifically — the pan-cancer DM1 vulnerability holds in thyroid-lineage cells. PAX8 and NKX2-1 are

> correspondingly less essential in DM1-high (d = +0.42 and +0.26) and only 9% essential in thyroid lineage — consistent with their lineage-TF role rather than housekeeping. Therapy candidates: BET inhibitors (MYC), FK866 (NAMPT), DNMT1-targeting hypomethylating agents (decitabine, azacitidine). The hypomethylating agent connection is particularly compelling given the fusion-independent promoter methylation finding (Result 2.7), suggesting decitabine + RAI re-induction trials (NCT00085293, NCT01065090) have a mechanistic rationale specifically in DM1-high patients.

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